

Programming Project: Phase - 1

ME – 609: Optimization Methods in Engineering

Group No: 18

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Introduction

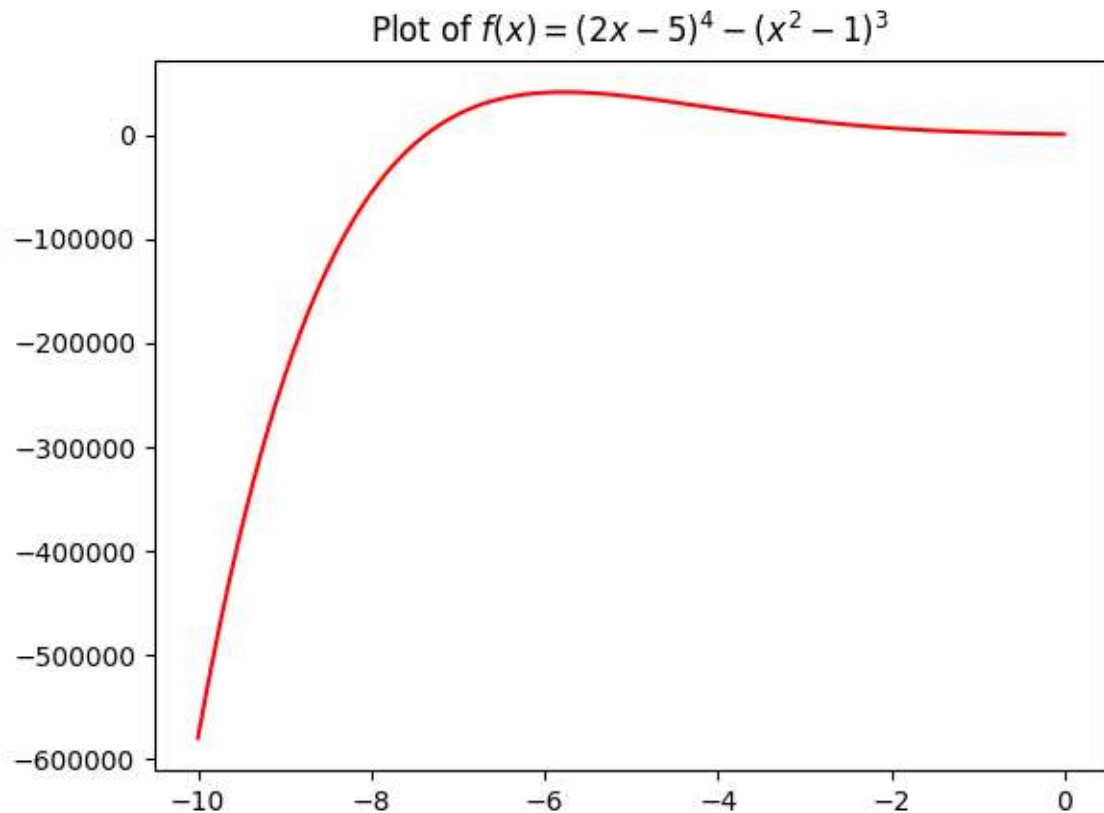
Objective: To solving 6 optimization problems using Bounded Phase method and Bisection Method.

For each of the six problem, we follow a approach:

1. Bounding Phase method to identify an initial interval containing the optimum
2. Bisection method: To refine the search and pinpoint the optimum more precisely.

For each problem, we perform multiple iterations with different random initial guesses.

Problem 1: Maximize $f(x) = (2x - 5)^4 + (x^2 - 1)^3$ in the interval $(-10, 0)$



$$f'(x) = 8(2x - 5)^3 - 6x(x^2 - 1)^2$$

$$f'(x) = 0 \Rightarrow x = -5.77507$$

$$f''(x) = 48(2x - 5)^2 - 24x(x^2 - 1) - 6(x^2 - 1)^2$$

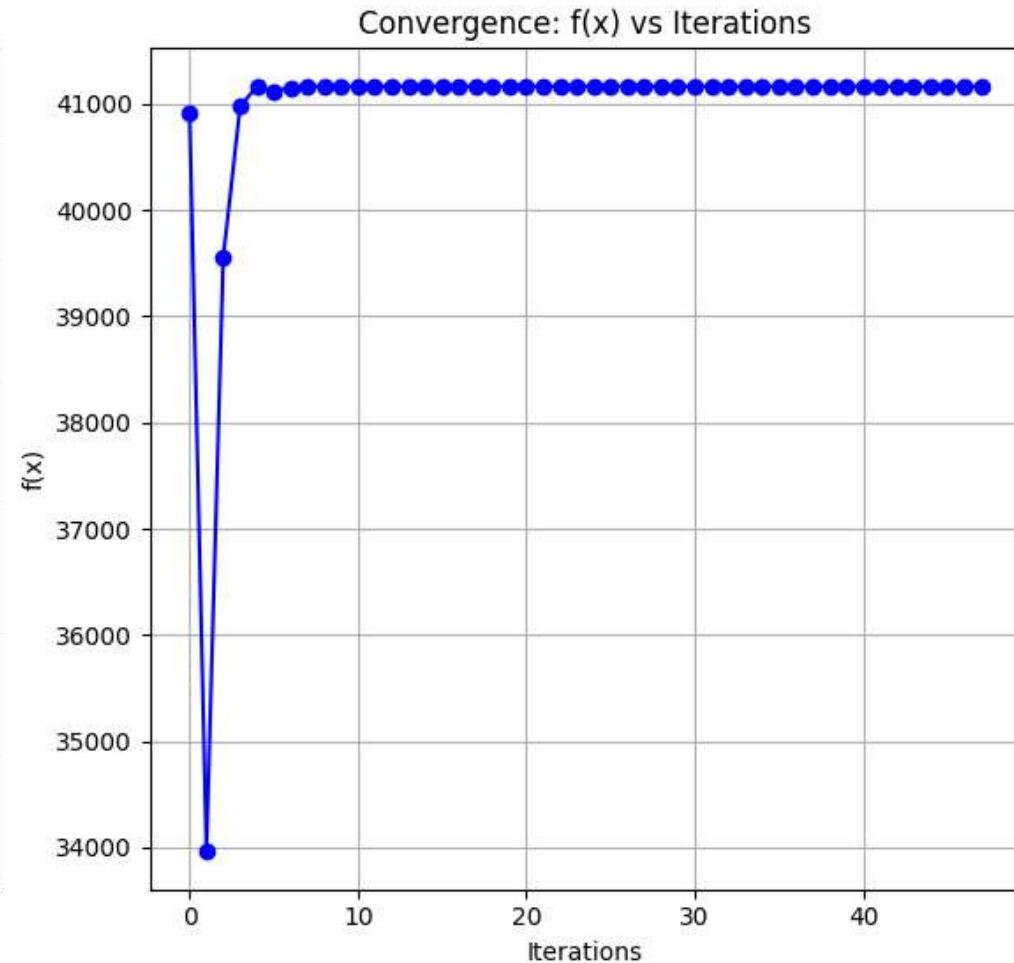
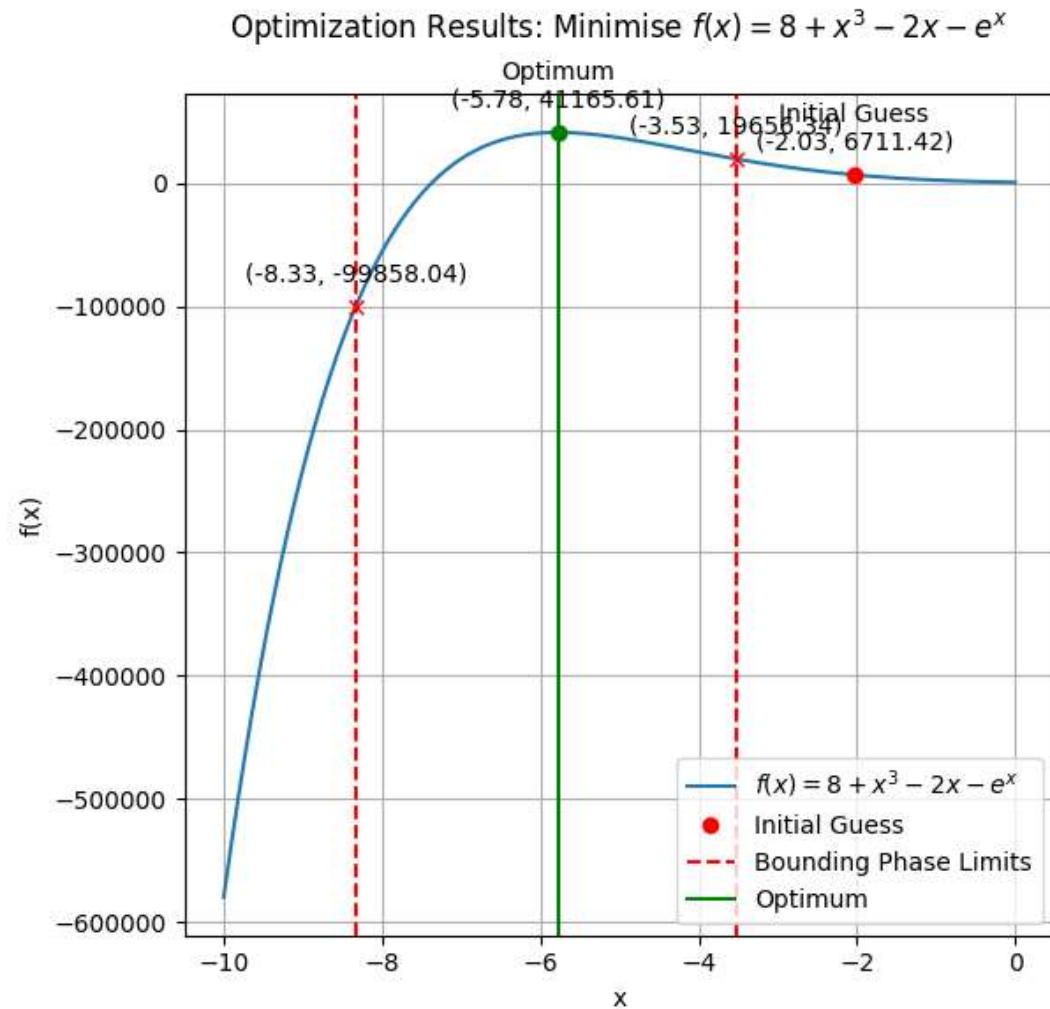
$$f''(-5.77507) = -19027.349 < 0$$

The maxima of the function exists at $x = -5.77507$
and optimum value is $f(x) = 41165.61$

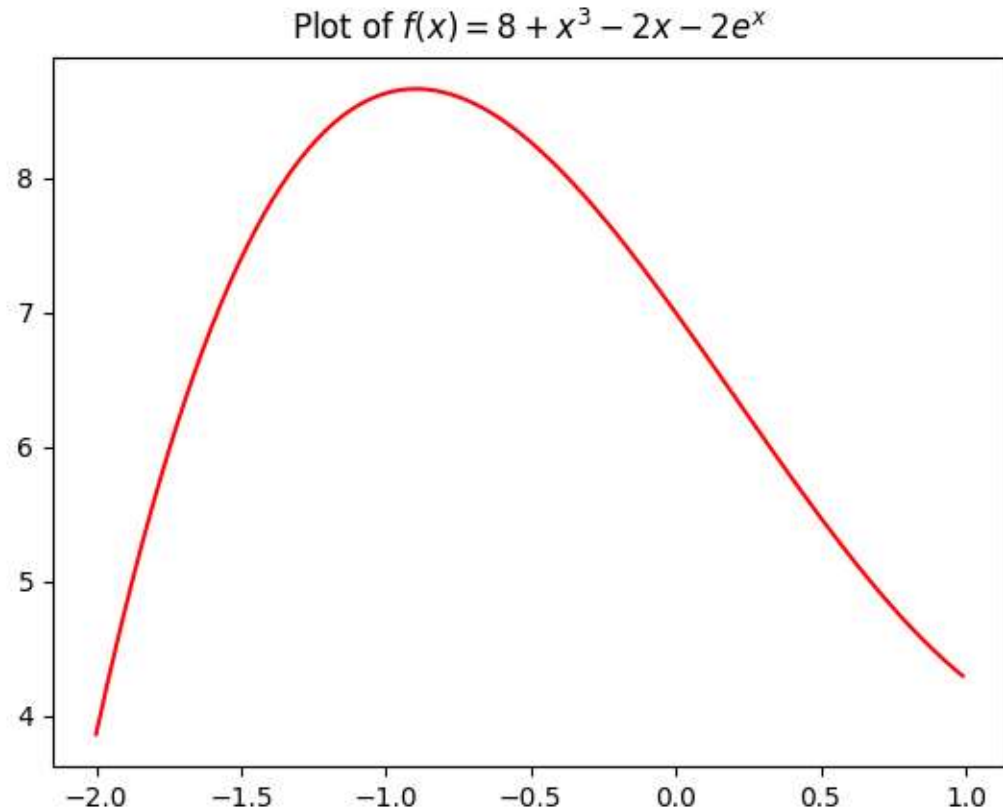
Problem 1: $f(x) = (2x - 5)^4 + (x^2 - 1)^3$ in the interval $(-10,0)$

S.No.	Initial Guess (x_0)	Approximate Solution (a, b)	Exact Solution (x_c)	Number of function evaluations
1.	-6.753	[-6.452,-5.252]	-5.7751	156
2.	-3.969	[-7.069,-4.669]	-5.7751	160
3.	-7.283	[-6.583,-4.183]	-5.7751	160
4.	-4.616	[-7.717,-5.317]	-5.7751	160
5.	-4.320	[-7.240,-4.840]	-5.7751	160
6.	-7.160	[-6.460,-4.060]	-5.7751	160
7.	-4.891	[-6.391,-5.191]	-5.7751	114
8.	-2.030	[-8.330,-3.530]	-5.7751	152
9.	-3.601	[-6.701,-4.301]	-5.7751	160
10.	-0.284	[-12.984,-3.384]	-5.7751	168

Problem 1: $f(x) = (2x - 5)^4 + (x^2 - 1)^3$ in the interval $(-10,0)$



Problem 2: Maximize $f(x) = 8 + x^3 - 2x - 2e^x$ in the interval $(-2,1)$



$$f'(x) = 3x^2 - 2 - 2e^x$$

$$f'(x) = 0 \Rightarrow x = -0.96015$$

$$f''(x) = 6x - 2e^x$$

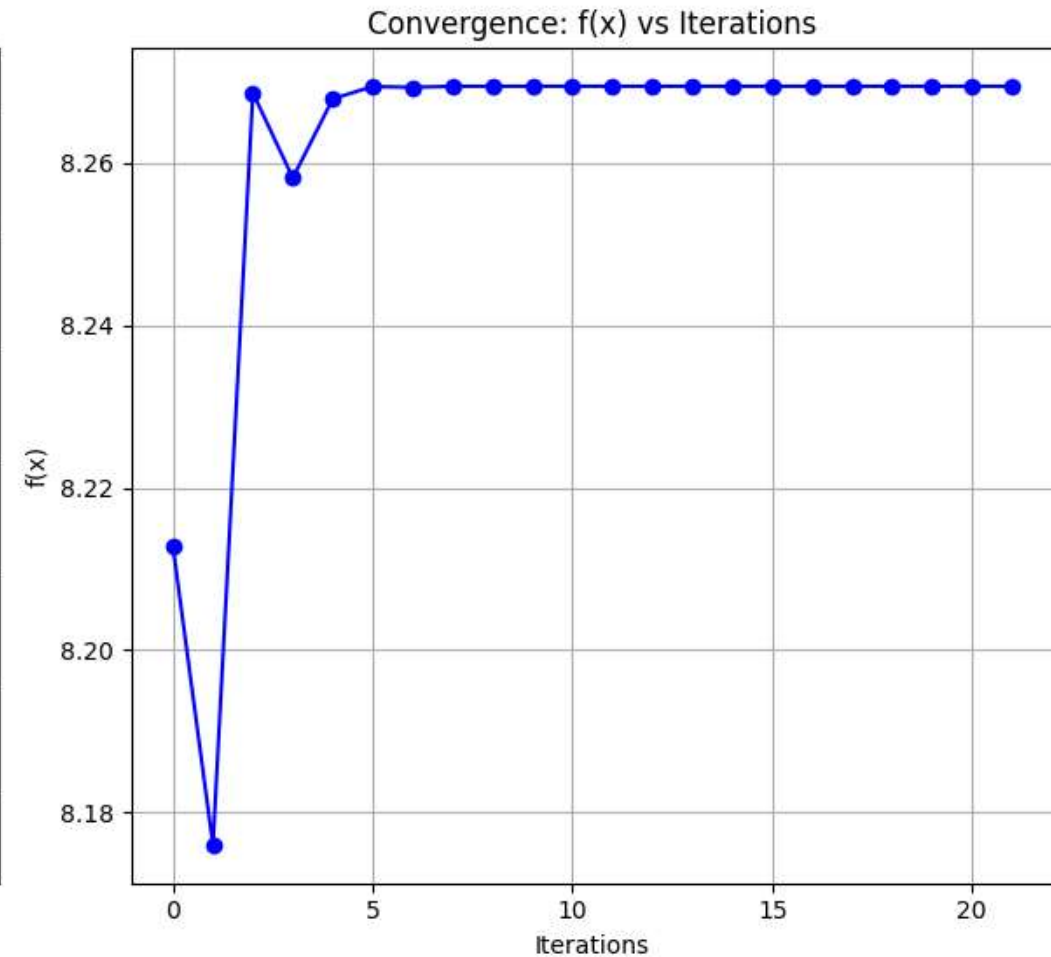
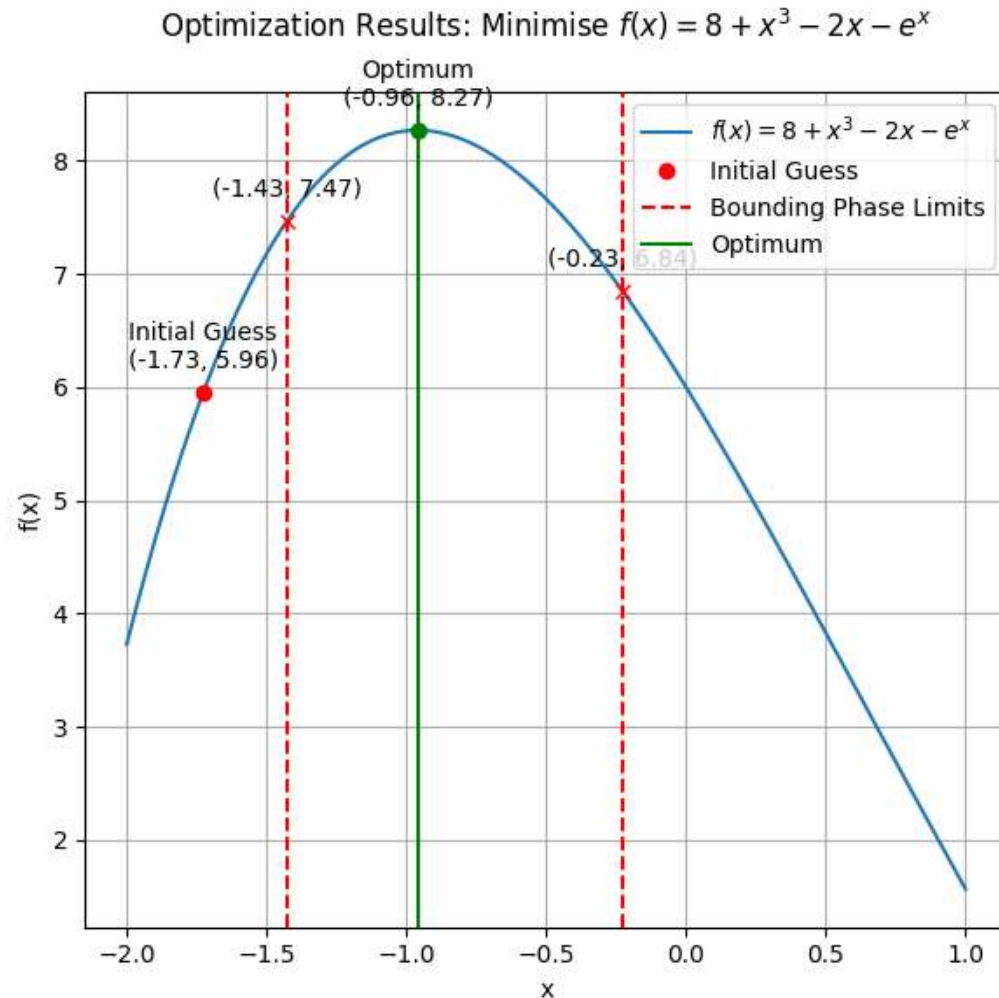
$$f''(-0.96015) = -6.5265 < 0$$

The maxima of the function exists at $x = -0.96015$
and optimum value is $f(x) = 8.2694$

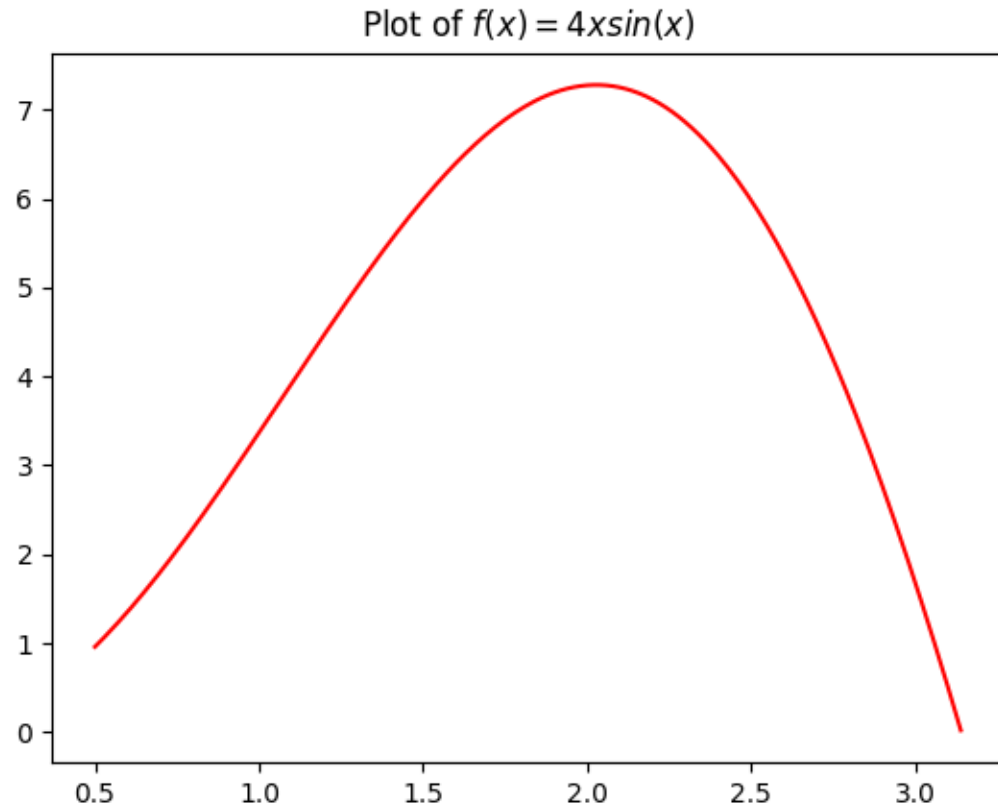
Problem 2: Maximize $f(x) = 8 + x^3 - 2x - 2e^x$ in the interval $(-2,1)$

S.No.	Initial Guess (x_0)	Approximate Solution (a, b)	Exact Solution (x_c)	Number of function evaluations
1.	0.707	[-2.392,0.007]	-0.9602	76
2.	0.912	[-2.187,0.212]	-0.9602	67
3.	-0.902	[-1.202,-9.024]	-0.9602	64
4.	-1.726	[-1.426,-0.226]	-0.9602	72
5.	-0.002	[-1.502,-0.302]	-0.9602	66
6.	0.756	[-2.344,0.556]	-0.9602	76
7.	-0.391	[-1.891,-0.691]	-0.9602	63
8.	-1.803	[-1.503,-0.303]	-0.9602	63
9.	-1.288	[-1.188,-0.588]	-0.9602	68
10.	0.750	[-2.341,0.059]	-0.9602	76

Problem 2: Maximize $f(x) = 8 + x^3 - 2x - 2e^x$ in the interval $(-2,1)$



Problem 3: Maximize $f(x) = 4x\sin(x)$ in the interval $(0.5, \pi)$



$$f'(x) = 4(x\cos(x) + \sin(x))$$

$$f'(x) = 0 \Rightarrow x = 2.02876$$

$$f''(x) = 4(-x\sin(x) + 2\cos(x))$$

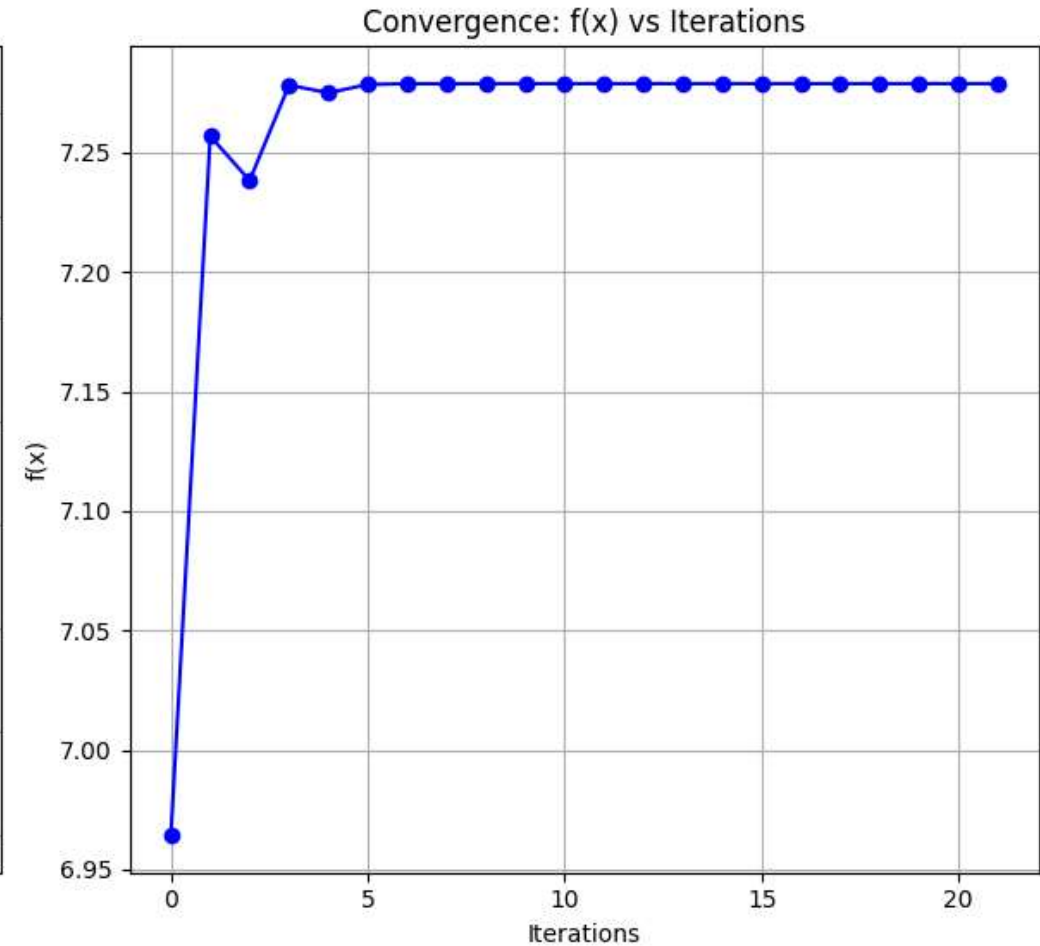
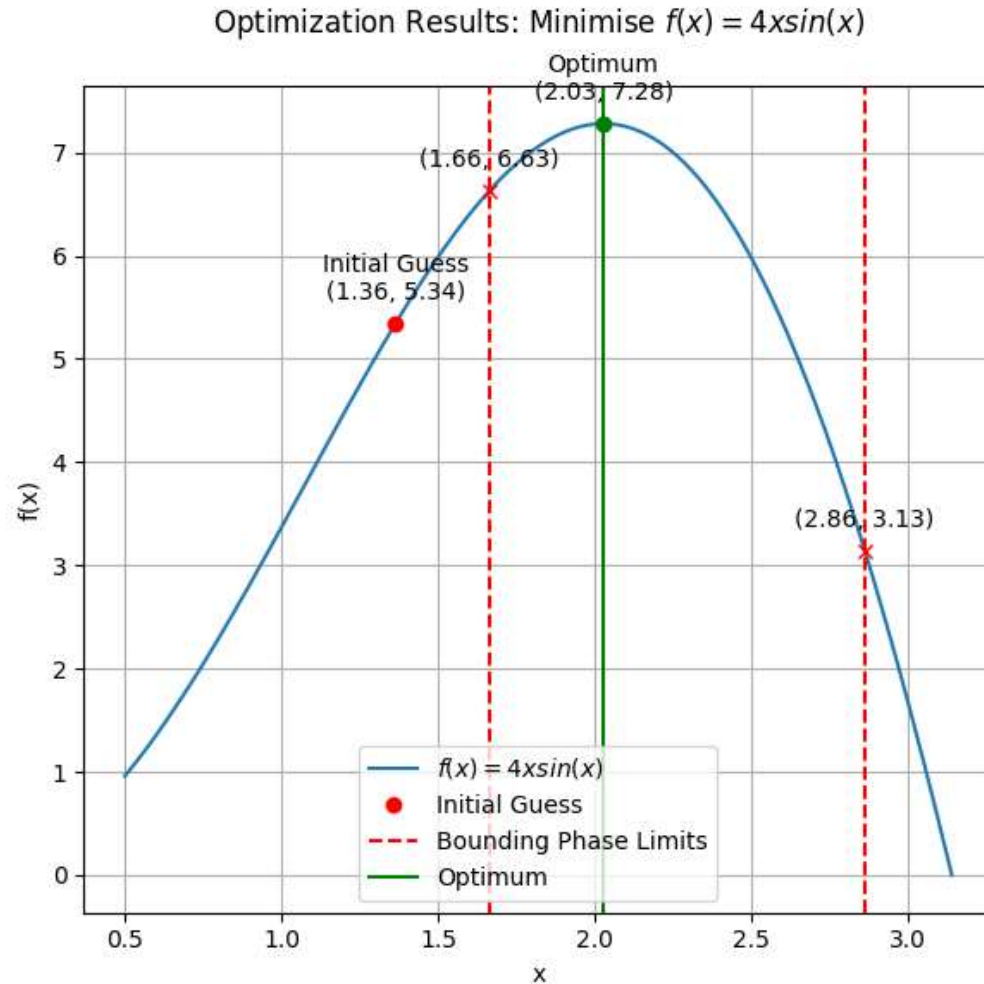
$$f''(2.02876) = -6.5265 < 0$$

The maxima of the function exists at $x = 2.02876$ and optimum value is $f(x) = -10.8158$

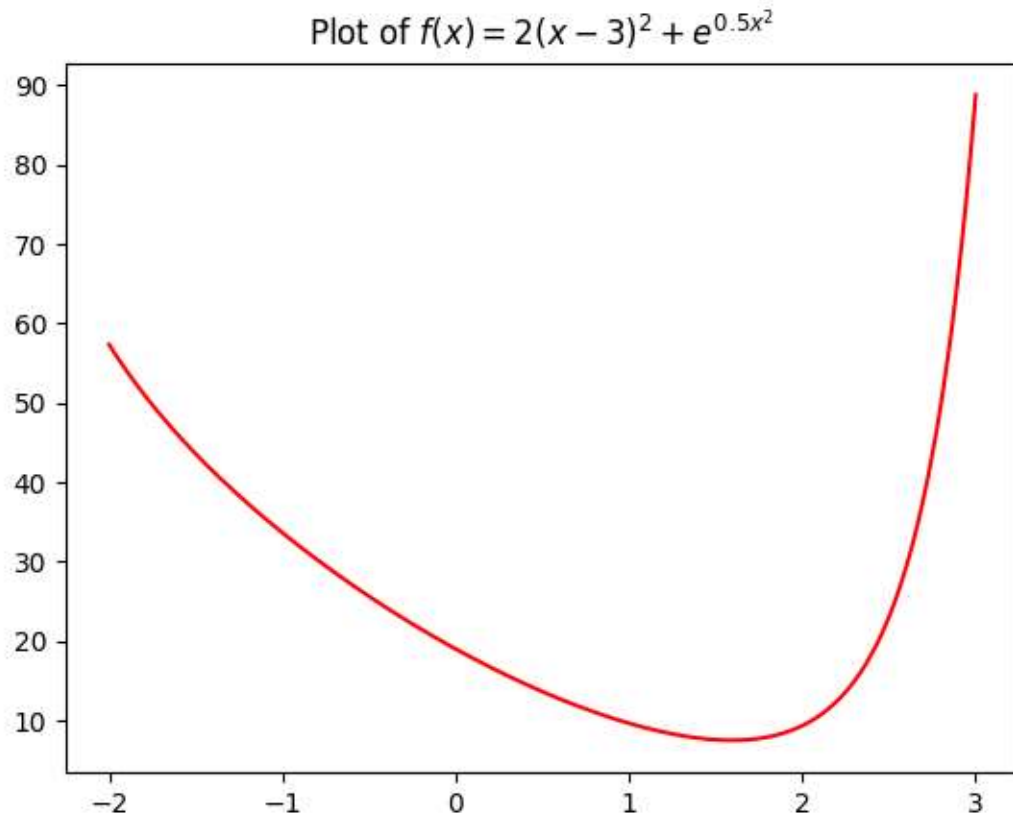
Problem 3: Maximize $f(x) = 4x\sin(x)$ in the interval $(0.5, \pi)$

S.No.	Initial Guess (x_0)	Approximate Solution (a, b)	Exact Solution (x_c)	Number of function evaluations
1.	1.364	[1.664,2.864]	2.0288	72
2.	1.025	[1.325,2.525]	2.0288	75
3.	0.662	[1.362,3.762]	2.0288	76
4.	0.831	[1.531,3.931]	2.0288	79
5.	2.717	[1.217,2.417]	2.0288	75
6.	3.055	[1.555,2.755]	2.0288	75
7.	1.290	[1.590,2.790]	2.0288	54
8.	1.202	[1.501,2.701]	2.0288	75
9.	1.112	[1.412,2.612]	2.0288	75
10.	3.122	[0.022,2.422]	2.0288	70

Problem 3: Maximize $f(x) = 4x\sin(x)$ in the interval $(0.5, \pi)$



Problem 4: Minimize $f(x) = 2(x - 3)^2 + e^{(0.5x^2)}$ in interval $(-2,3)$



$$f'(x) = xe^{\frac{x^2}{2}} + 4(x - 3)$$

$$f'(x) = 0 \Rightarrow x = 1.59072$$

$$f''(x) = (x^2 + 1)e^{\frac{x^2}{2}} + 4$$

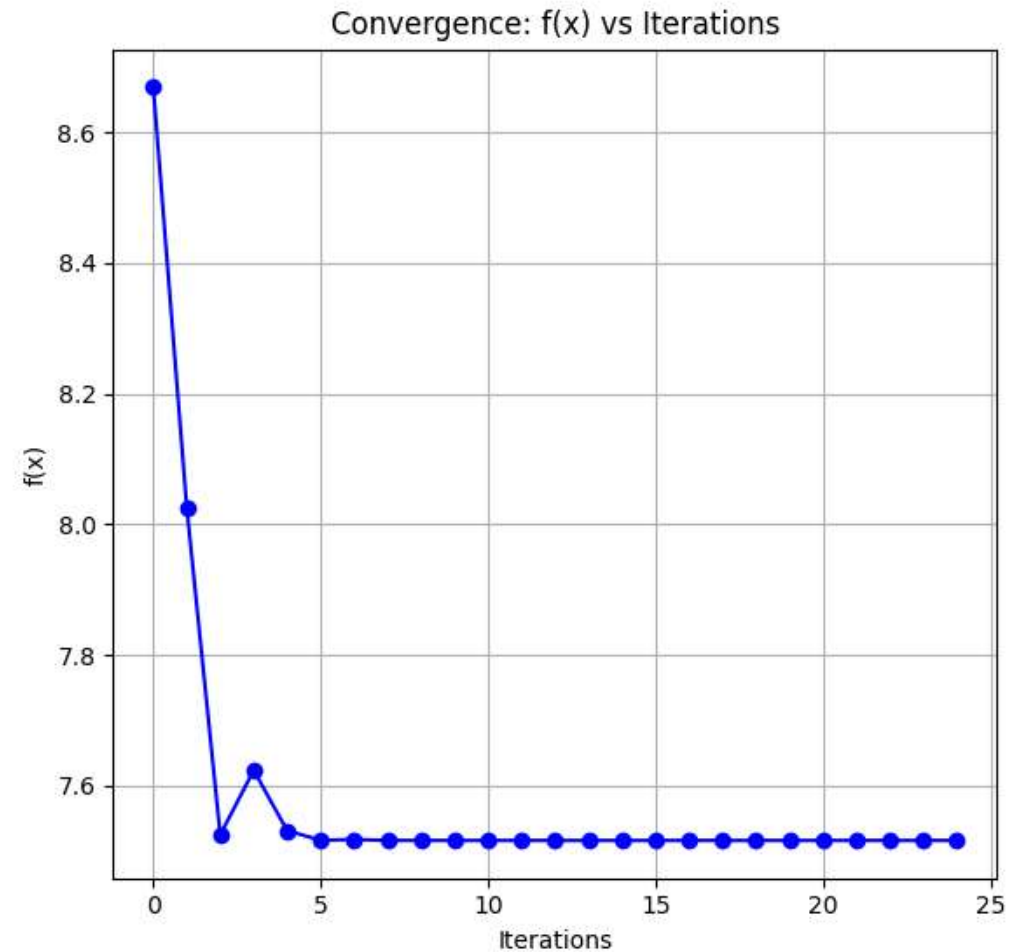
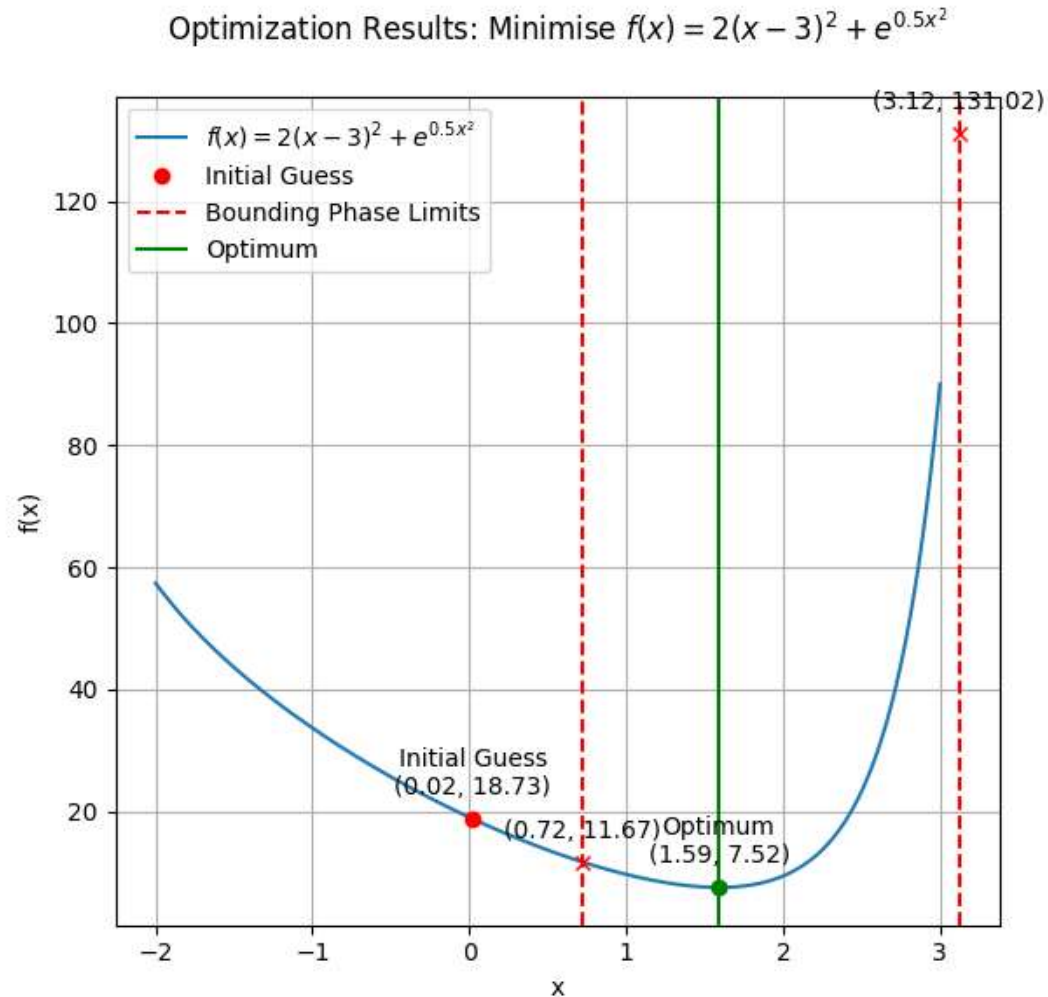
$$f''(1.59072) = 91.3706 > 0$$

The minima of the function exists at $x = 1.59072$ and optimum value is $f(1.59072) = 7.51592$

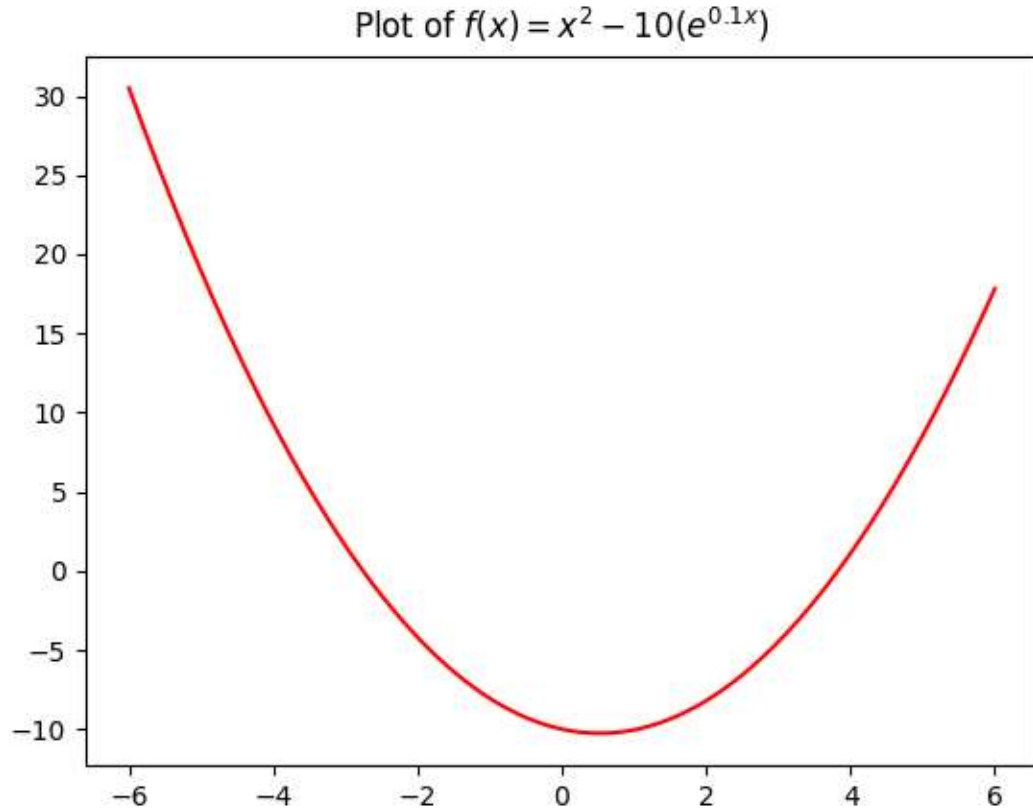
Problem 4: Minimize $f(x) = 2(x - 3)^2 + e^{(0.5x^2)}$ in interval $(-2,3)$

S.No.	Initial Guess (x_0)	Approximate Solution (a, b)	Exact Solution (x_c)	Number of function evaluations
1.	-0.225	[0.722, 3.122]	1.59	59
2.	2.69	[-0.41, 1.99]	1.59	58
3.	2.34	[0.843, 2.043]	1.59	57
4.	2.76	[-0.343, 2.058]	1.59	58
5.	0.60	[1.00, 2.20]	1.59	58
6.	2.41	[0.91, 2.11]	1.59	57
7.	2.24	[0.74, 1.94]	1.59	57
8.	0	[0.8, 3.2]	1.59	59
9.	-0.96	[0.16, 2.24]	1.59	59
10.	-0.92	[-0.12, 2.28]	1.59	59

Problem 4: Minimize $f(x) = 2(x - 3)^2 + e^{(0.5x^2)}$ in interval $(-2,3)$



Problem 5: Minimize $f(x) = x^2 - 10e^{0.1x}$ in interval $(-6,6)$



$$f'(x) = 2x - e^{\frac{x}{10}}$$

$$f'(x) = 0 \Rightarrow x = 0.526$$

$$f''(x) = 2 - 0.1e^{0.1x}$$

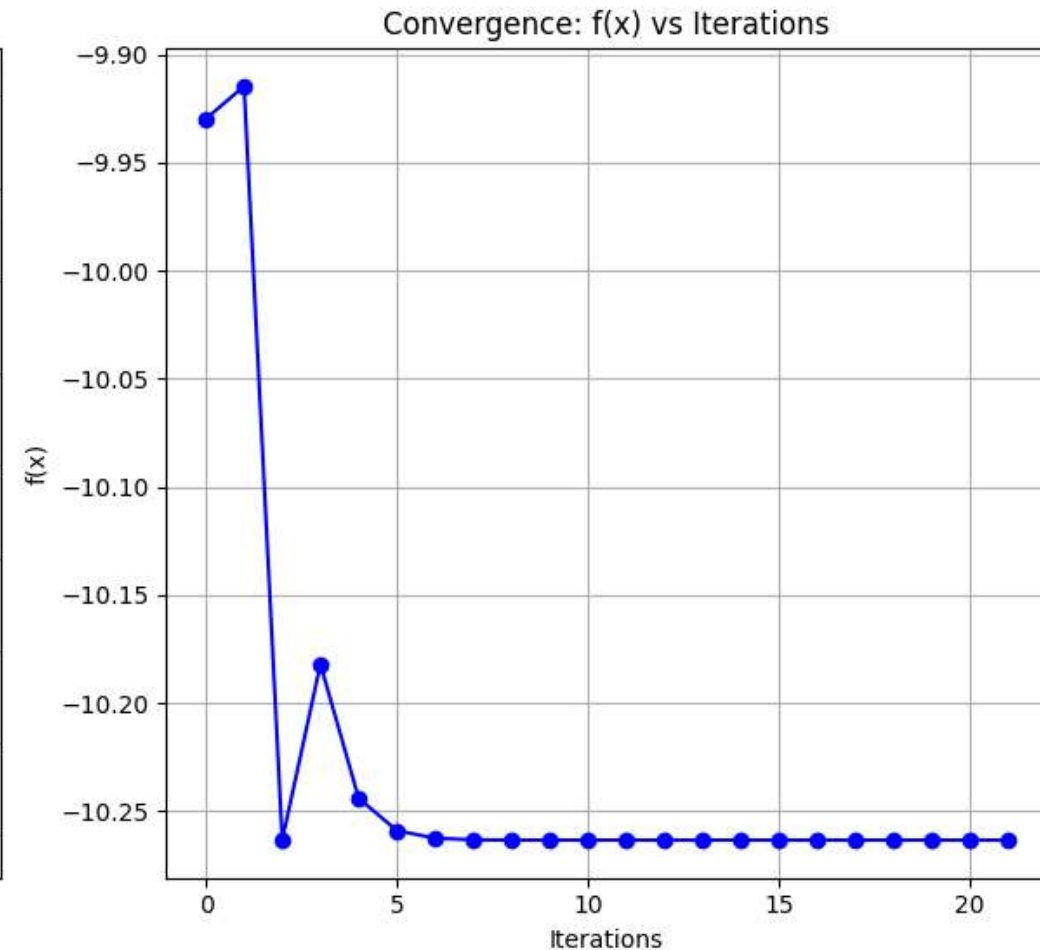
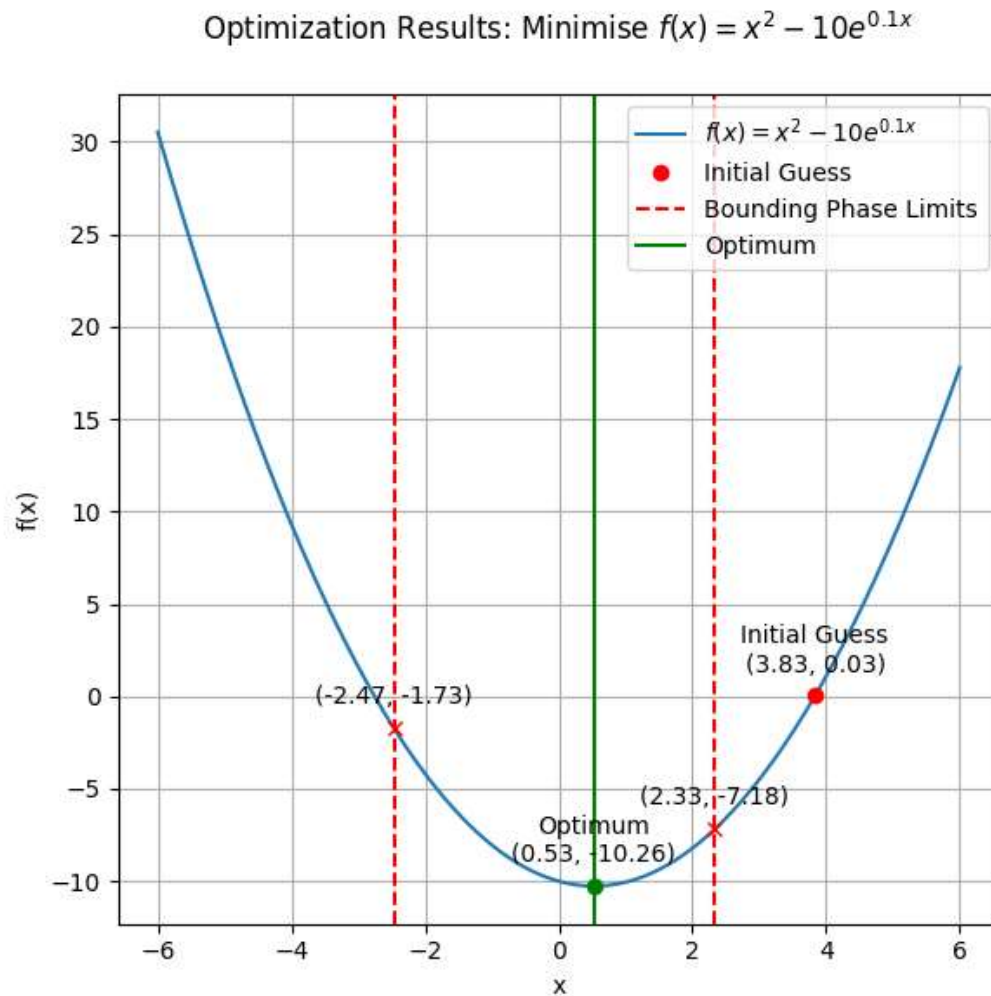
$$f''(0.526) = 1.8496 > 0$$

The minima of the function exists at $x = 0.526$ and optimum value is $f(0.526) = -10.2634$

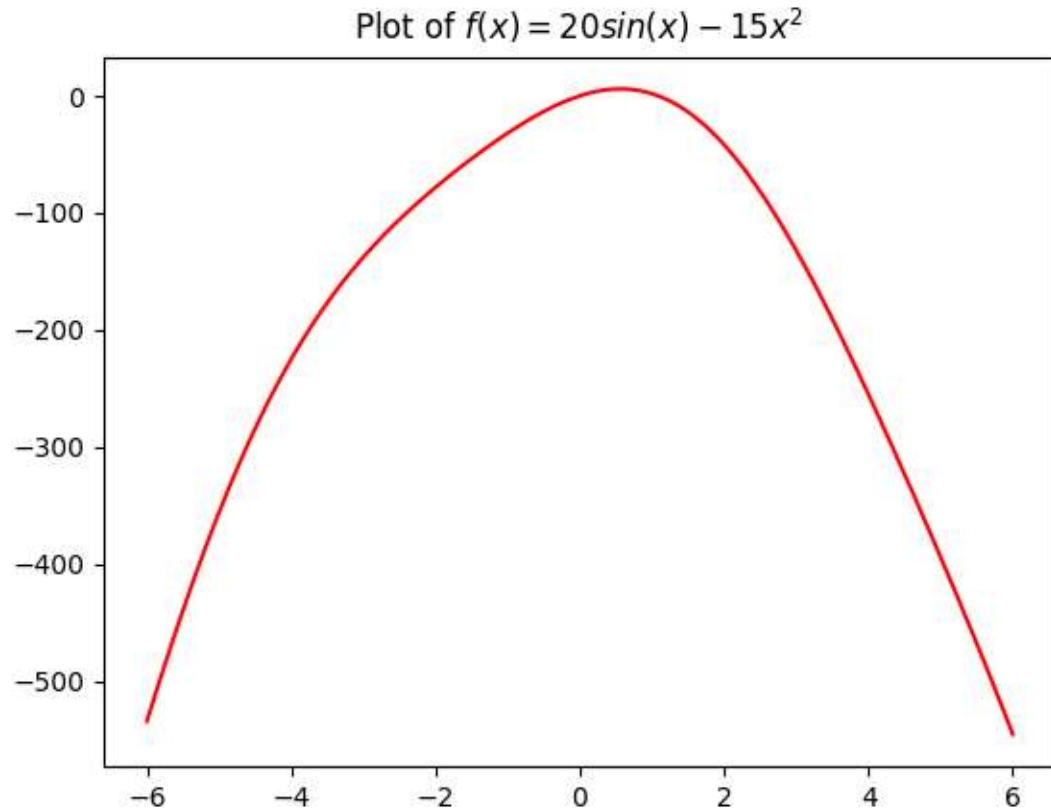
Problem 5: Minimize $f(x) = x^2 - 10e^{0.1x}$ in interval $(-6,6)$

S.No.	Initial Guess (x_0)	Approximate Solution (a, b)	Exact Solution (x_c)	Number of function evaluations
1.	4.14	$[-2.16, 2.64]$	0.53	61
2.	2.90	$[-3.40, 1.40]$	0.53	61
3.	-3.43	$[-1.83, 2.97]$	0.53	62
4.	-3.52	$[-1.92, 2.88]$	0.53	62
5.	-5.91	$[-2.71, 6.89]$	0.53	63
6.	2.44	$[-0.66, 1.74]$	0.53	60
7.	3.59	$[-2.71, 2.09]$	0.53	61
8.	1.45	$[-0.04, 1.15]$	0.53	66
9.	3.55	$[-2.75, 2.05]$	0.53	61
10.	3.07	$[-3.23, 1.57]$	0.53	61

Problem 5: Minimize $f(x) = x^2 - 10e^{0.1x}$ in interval $(-6,6)$



Problem 6: Maximize $f(x) = 20 \sin x - 15x^2$ in interval $(-4,4)$



$$f'(x) = 20 \cos x - 30x$$

$$f'(x) = 0 \Rightarrow x = 0.564$$

$$f''(x) = -20 \sin x - 30$$

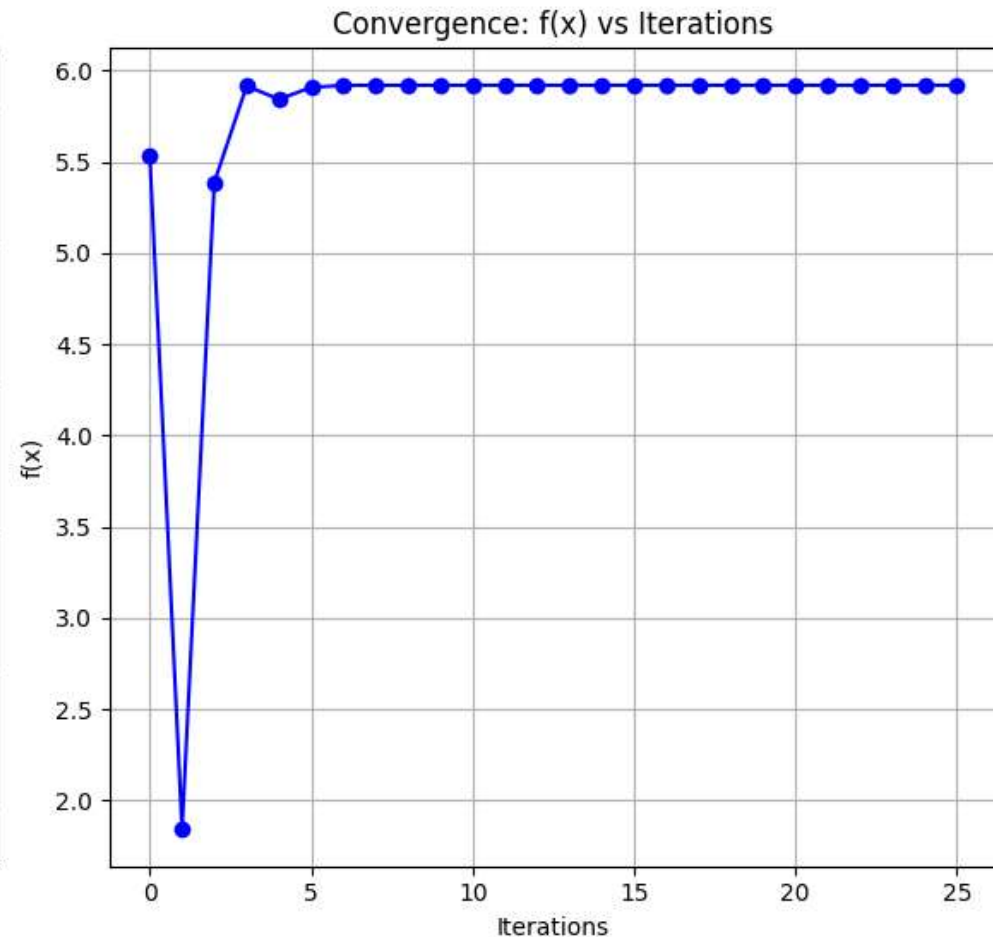
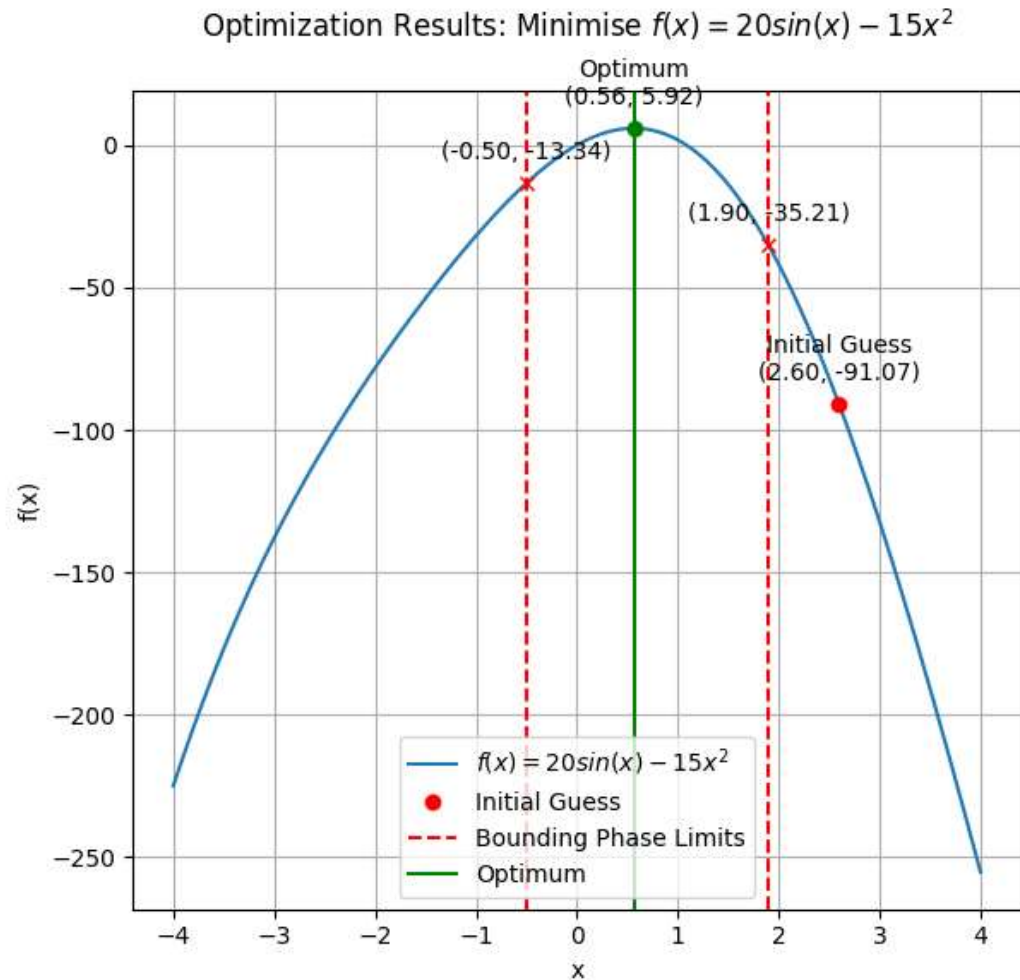
$$f''(0.564) = -40.6914 < 0$$

The maxima of the function exists at $x = 0.564$ and optimum value is $f(0.564) = 5.91998$

Problem 6: Maximize $f(x) = 20 \sin x - 15x^2$ in interval $(-4,4)$

S.No.	Initial Guess (x_0)	Approximate Solution (a, b)	Exact Solution (x_c)	Number of function evaluations
1.	3.68	$[-2.62, 2.18]$	0.56	67
2.	-3.15	$[-1.55, 3.25]$	0.56	66
3.	-2.14	$[-0.54, 4.26]$	0.56	68
4.	3.94	$[-2.36, 2.44]$	0.56	67
5.	-3.3	$[-1.70, 3.10]$	0.56	66
6.	3.27	$[-3.03, 1.77]$	0.56	67
7.	-2.08	$[-0.48, 4.32]$	0.56	68
8.	3.81	$[-2.49, 2.31]$	0.56	67
9.	-2.43	$[-0.83, 3.97]$	0.56	66
10.	2.60	$[-0.50, 1.90]$	0.56	65

Problem 6: Maximize $f(x) = 20 \sin x - 15x^2$ in interval $(-4,4)$



Conclusions

- With different values of initial guess, we get the same minima or maxima value
- Lesser value of delta will give a shorter bound but, the number of iterations will be larger.

Programming Project: Phase - 2

ME – 609: Optimization Methods in Engineering

Group No: 18

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Introduction

Objective: To solving 5 optimization problems using Conjugate Gradient Method.

Conjugate Gradient Method:

$$s^k = -\nabla f(x^k) + \frac{\|\nabla f(x^k)\|^2}{\|\nabla f(x^{k-1})\|^2} s^{k-1}$$

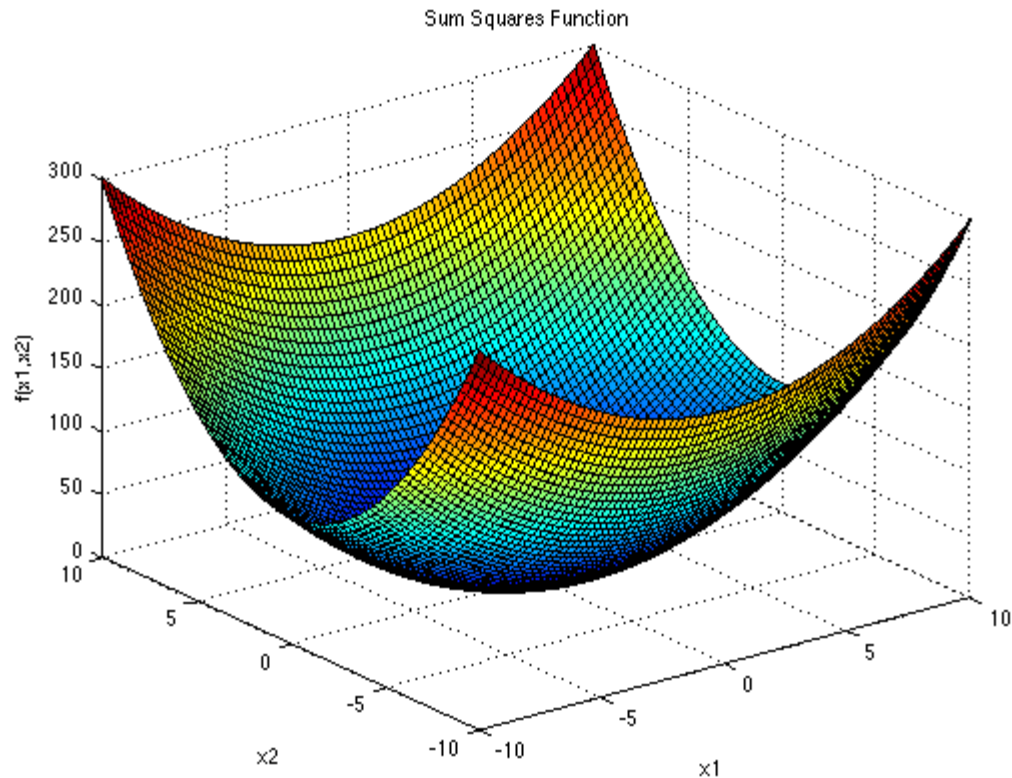
Termination Parameters:

For Unidirectional Search: Checking Linear dependency.

Also, Restart condition is introduced for code to have better convergence and accurate results.

Problem 1: Sum Square Function (for 5 variables)

$$f(x) = \sum_{i=1}^{n=5} i \cdot x_i^2$$



Characteristics:

- Convex: Globally convex, making it easier to optimize
- Unimodal: Has a single minimum
- Separability: Each variable is independent:

Global Minimum:

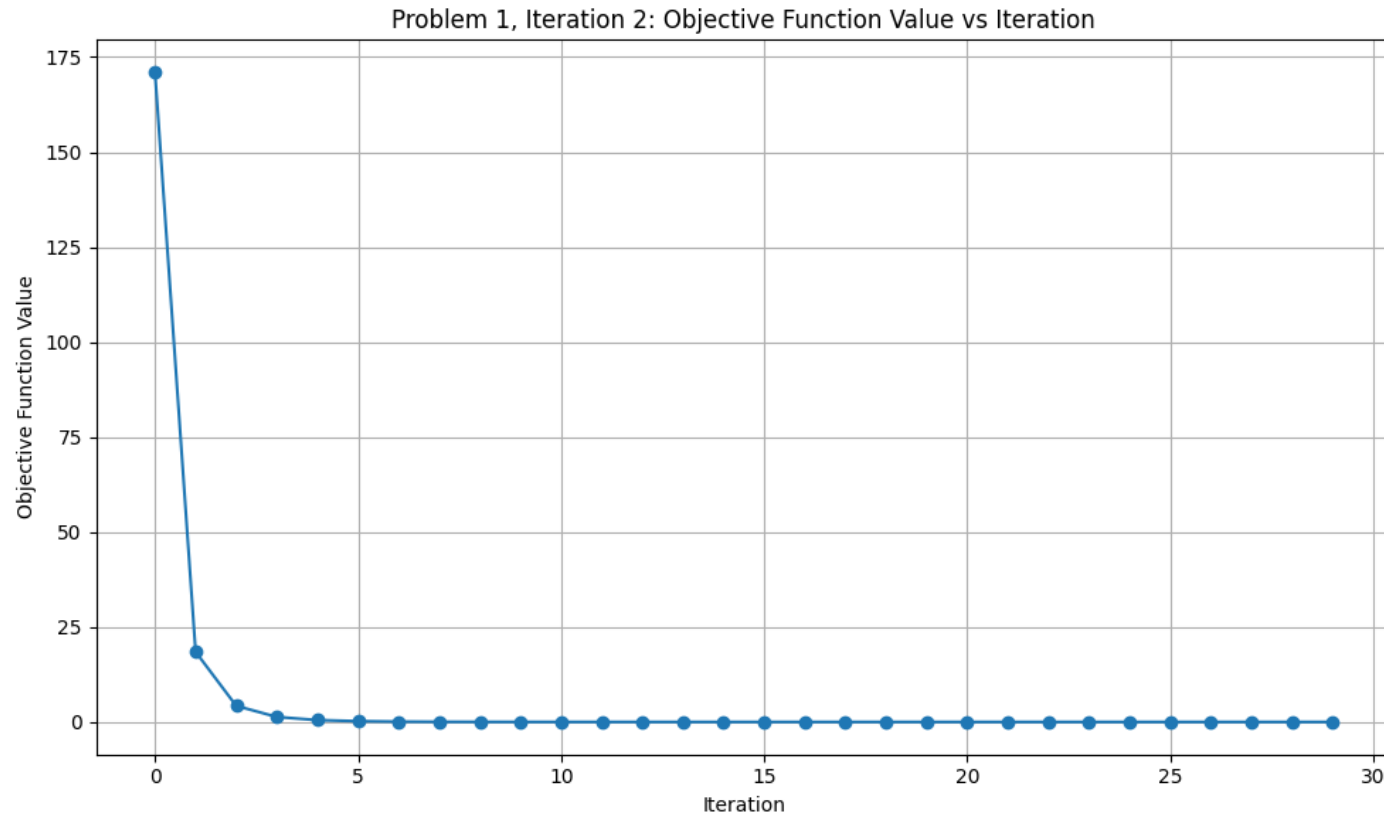
- Occurs at $x = [0,0,0,0,0]$
- $f(x) = 0$

Nature: Parabolic and simple to optimize

Problem 1: Sum Square Function (for 5 variables)

S.No.	Initial Guess (x_0)	Exact Solution (x_c)	Number of Iterations	Function Evaluations
1	[0.772,-3.851,3.079,-3.779,-3.952]	[0.0, 0.0, 0.0, 0.0, 0.0]	27	358
2	[2.696,-4.924,-0.308,4.441,2.686]	[0.0, 0.0, 0.0, 0.0, 0.0]	29	432
3	[-1.368,3.333,-2.952,-2.645,3.788]	[0.0, 0.0, 0.0, 0.0, 0.0]	28	395
4	[2.156,2.521,2.711,-4.445,2.078]	[0.0, 0.0, 0.0, 0.0, 0.0]	29	405
5	[4.872,-1.332,-1.169,3.907,-4.136]	[0.0, 0.0, 0.0, 0.0, 0.0]	31	449
6	[2.914,0.926,4.180,0.535,-0.375]	[0.0, 0.0, 0.0, 0.0, 0.0]	28	392
7	[3.120,-2.226,2.440,2.796,5.082]	[0.0, 0.0, 0.0, 0.0, 0.0]	29	417
8	[-3.193,3.701,2.968,5.046,-1.631]	[0.0, 0.0, 0.0, 0.0, 0.0]	31	420
9	[2.763,0.184,-3.732,3.893,4.276]	[0.0, 0.0, 0.0, 0.0, 0.0]	29	414
10	[0.717,-3.850,3.052,-3.810,-3.952]	[0.0, 0.0, 0.0, 0.0, 0.0]	27	345

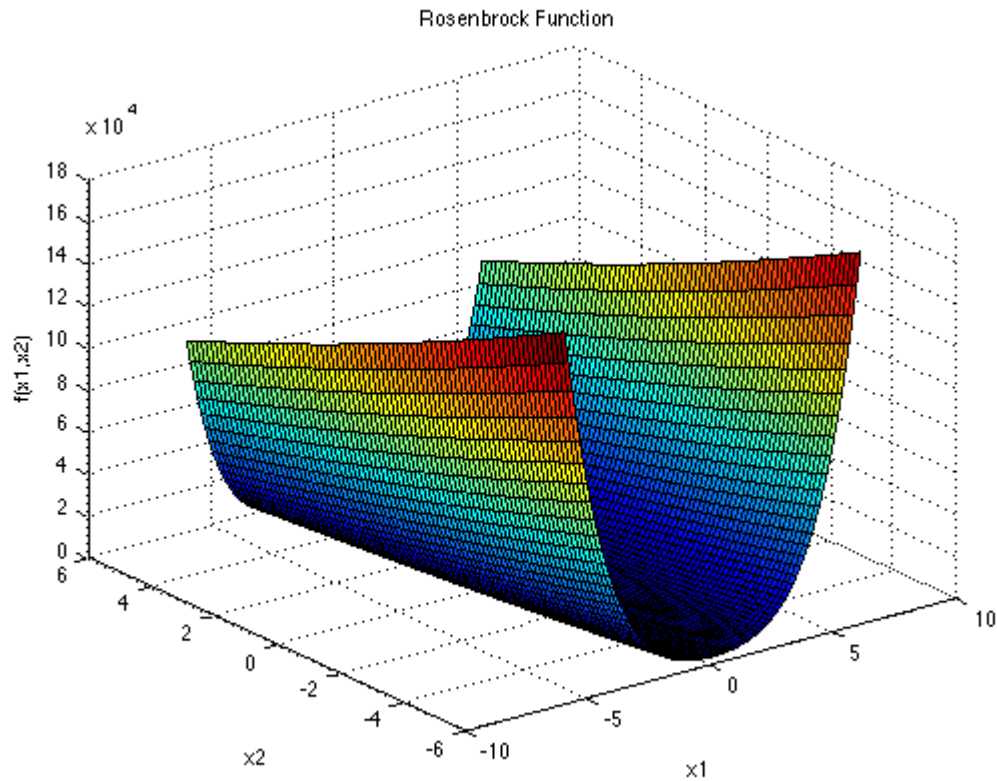
Problem 1: Sum Square Function (for 5 variables)



- The minima is found to be when all variables are 0 as expected
- The Objective function is 0 as expected

Problem 2: Rosenbrock Function (for 3 variables)

$$f(x) = \sum_{i=1}^2 [100(x_{i+1} - x_i^2)^2 + (1 - x_i)^2]$$



Characteristics:

- Non-convex: A valley-shaped function with a narrow, curved minimum.
- Unimodal: One global minimum, but difficult to find due to narrow valley.
- Gradient: High sensitivity to small changes in x .

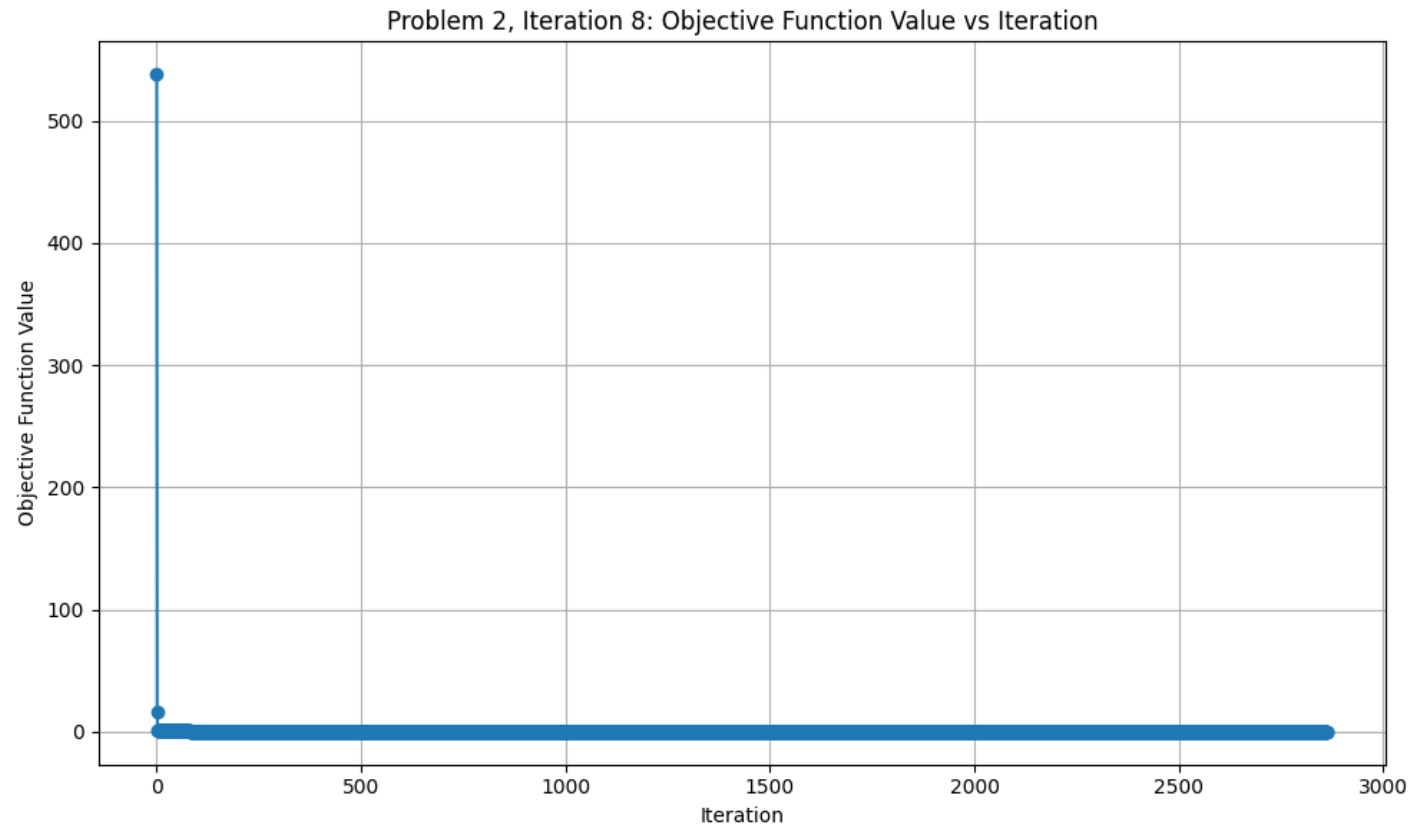
Global minimum:

- Occurs at $x = [1, 1, 1]$
- $f(x) = 0$

Problem 2: Rosenbrock Function (for 3 variables)

S.No.	Initial Guess (x_0)	Exact Solution (x_c)	Number of Iterations	Function Evaluations
1	[1.737,-1.095,0.749]	[0.9997,0.9993,0.9986]	2400	11932
2	[0.104,-0.576,-1.242]	[0.9999,0.9998,0.9996]	1785	9692
3	[-0.444,1.036,1.538]	[0.9996,0.9992,0.9984]	4606	22853
4	[1.216,-1.738,-1.528]	[0.9994,0.9989,0.9977]	2716	14156
5	[-1.972,1.369,-1.733]	[1,0.9999,0.9999]	2966	11858
6	[1.078,1.852,-1.540]	[0.9998,0.9997,0.9993]	2208	13261
7	[1.623,0.924,-1.869]	[1.0002,1.0004,1.0007]	1531	11583
8	[1.495,1.251,-0.534]	[0.9998,0.9995,0.9999]	2863	14530
9	[0.374,-1.042,0.056]	[0.9994,0.9989,0.9978]	2350	12976
10	[1.615,0.875,-1.674]	[0.9998,0.9999,0.9996]	2975	15351

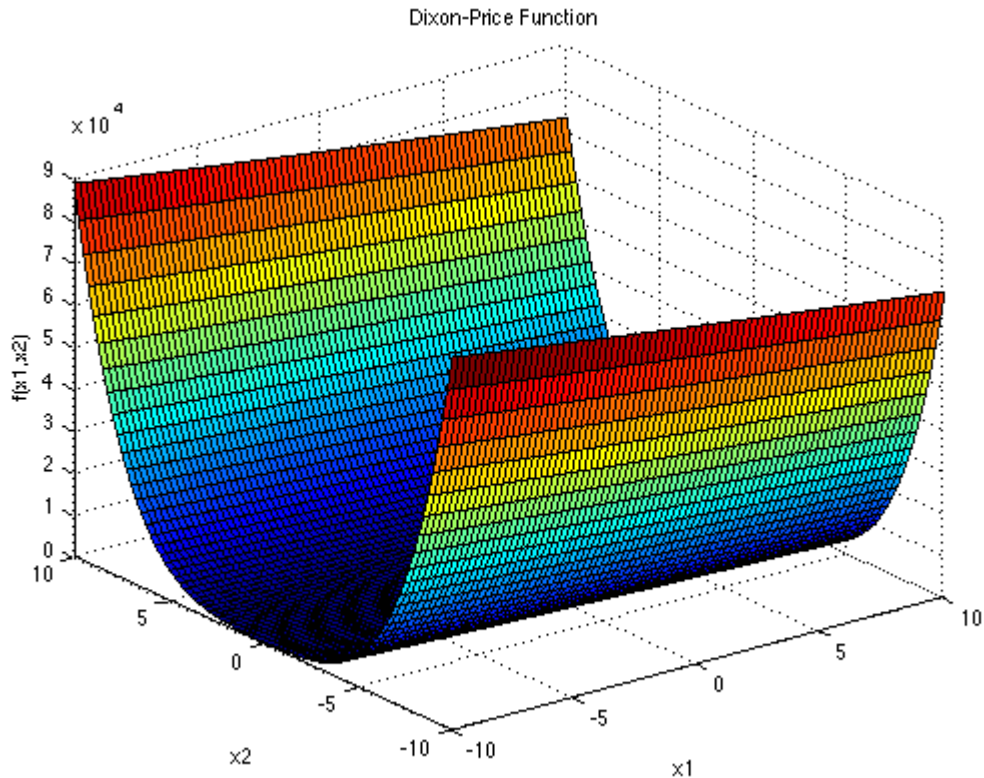
Problem 2: Rosenbrock Function (for 3 variables)



- The minima is found to be when all variables are 1 as expected
- The Objective function is 0 as expected

Problem 3: Dixon – Price Function (for 4 variables)

$$f(x) = (x_1 - 1)^2 + \sum_{i=1}^4 i[2x_i^2 - x_{i-1}]$$



Characteristics:

- Non-convex: Highly rugged landscape with many local minima
- Multimodal: Multiple local minima, but a clear global minimum
- Gradient: Complex interactions between variables

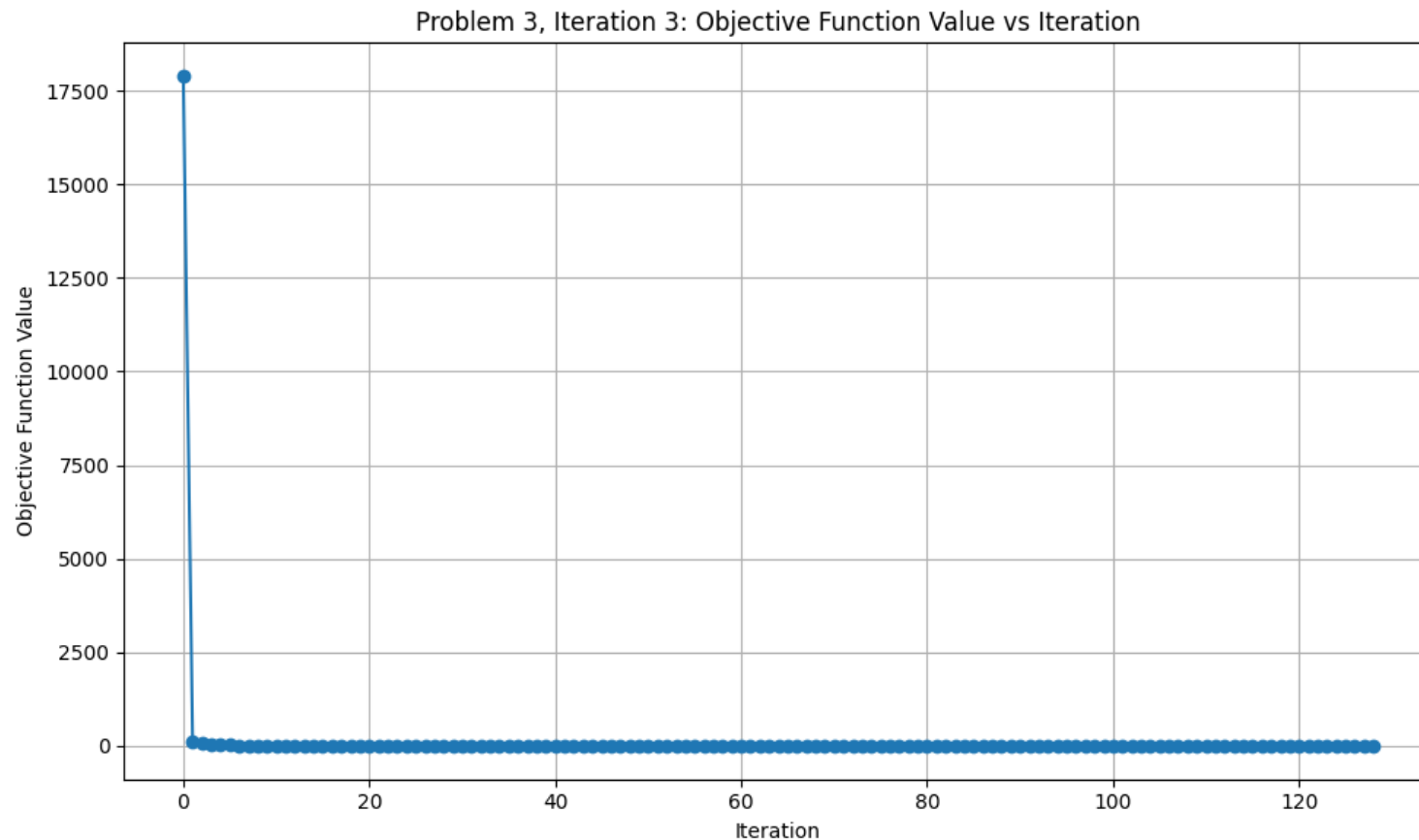
Global minimum:

- Occurs at $x = \left[1, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{3}, \frac{\sqrt{4}}{4}\right]$
- $f(x) = 0$

Problem 3: Dixon – Price Function (for 4 variables)

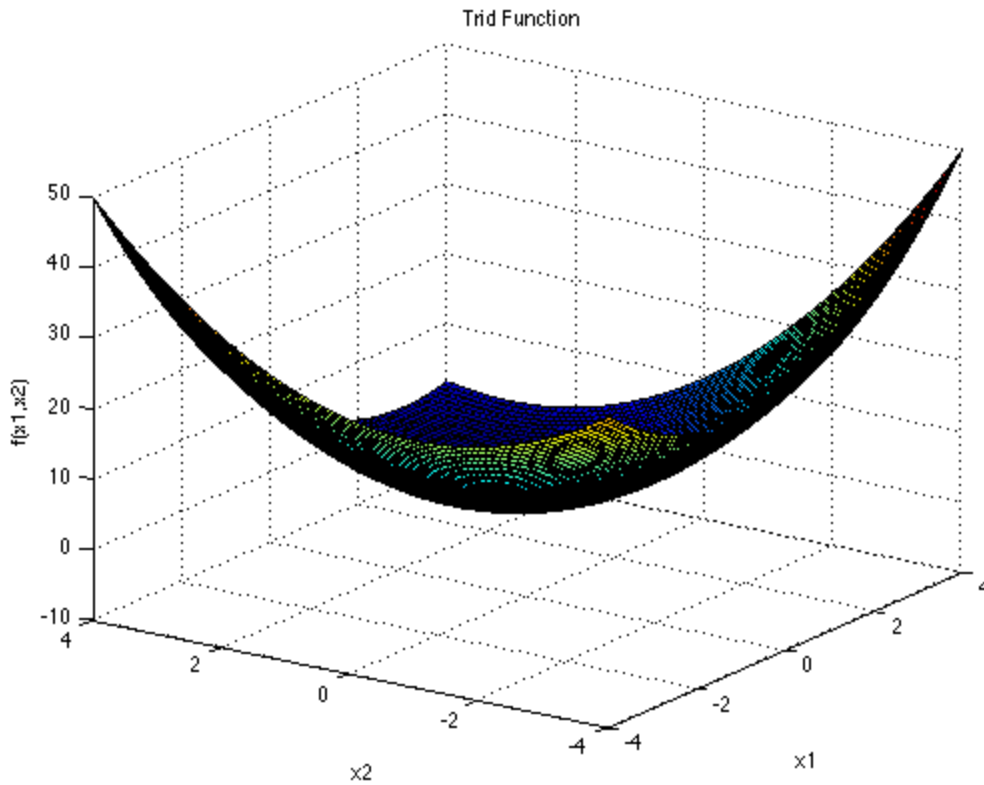
S.No .	Initial Guess (x_0)	Exact Solution (x_c)	Optimal Value	No of iterations	Function Evaluations
1	[-6.937,5.079 ,8.509,-8.429]	[0.3333,0,0,-0.0047]	0.6667	51	650
2	[-0.032,8.047,-6.025,9.822]	[1,0.707,0.5946,0.5453]	0	137	1454
3	[-5.117,-0.966,-5.578,4.026]	[1,0.707,0.5946,0.5453]	0	128	1776
4	[6.7,4.369,-6.231,-6.996]	[1,0.707,0.5946,0.5453]	0	145	2445
5	[-9.609,3.149,6.325,-3.619]	[3.333e-01,2.000e-04,3.300e-03, 4.090e-02]	0.6667	53	859
6	[-9.558,-0.012,-4.717,4.239]	[1,0.707,0.5946,-0.5453]	0	511	2631
7	[-6.27,7.821,4.203,3.527]	[1,0.707,0.5946,0.5453]	0	416	2637
8	[7.283,4.665,4.481,-4.654]	[1,0.707,0.5946,0.5453]	0	214	3389
9	[8.648,2.63,0.502,6.479]	[1,0.707,0.5946,-0.5453]	0	198	3477
10	[-4.191,2.44,0.654,-6.439]	[1,0.707,0.5946,-0.5453]	0	149	1954

Problem 3: Dixon – Price Function (for 4 variables)



- The minima is found to be when all variables are 0, but only for some cases
- The Objective function is 0 as expected only for some cases.
- Possible Reasons:
 - Non Convex Function
 - Usage of bisection method (Any value based method like golden section search could have given better results)

Problem 4: Trid Function (for 6 variables)



$$f(x) = \sum_{i=1}^6 (x_i - 1)^2 + \sum_{i=2}^6 x_i x_{i-1}$$

Characteristics:

- Non-convex: Exhibits significant non-linearity.
- Multimodal: Many local minima.
- Complexity: Variables are interconnected, making gradient descent more difficult

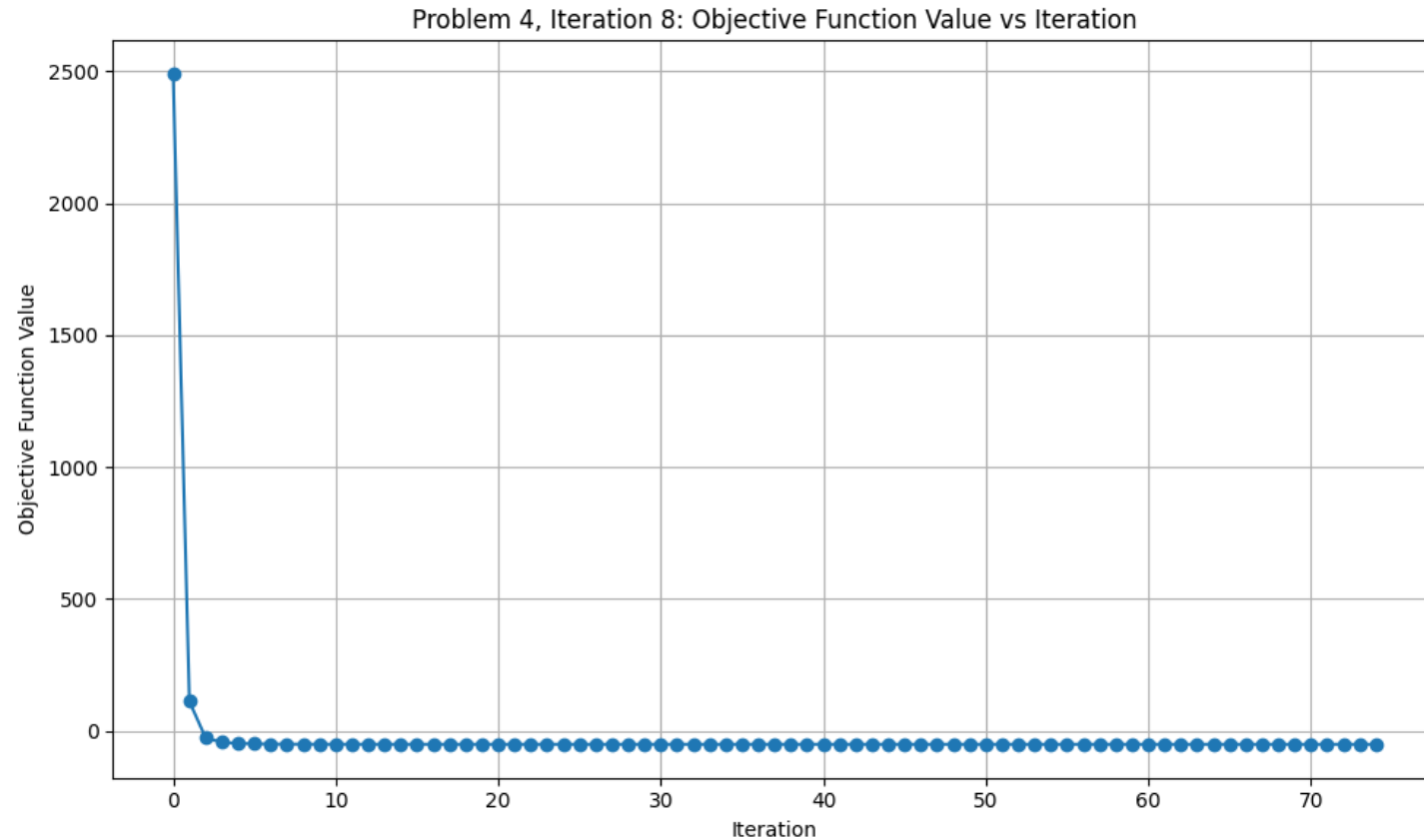
Global minimum:

- Occurs at $x = [6, 10, 12, 12, 10, 6]$
- $f(x) = -50$

Problem 4: Trid Function (for 6 variables)

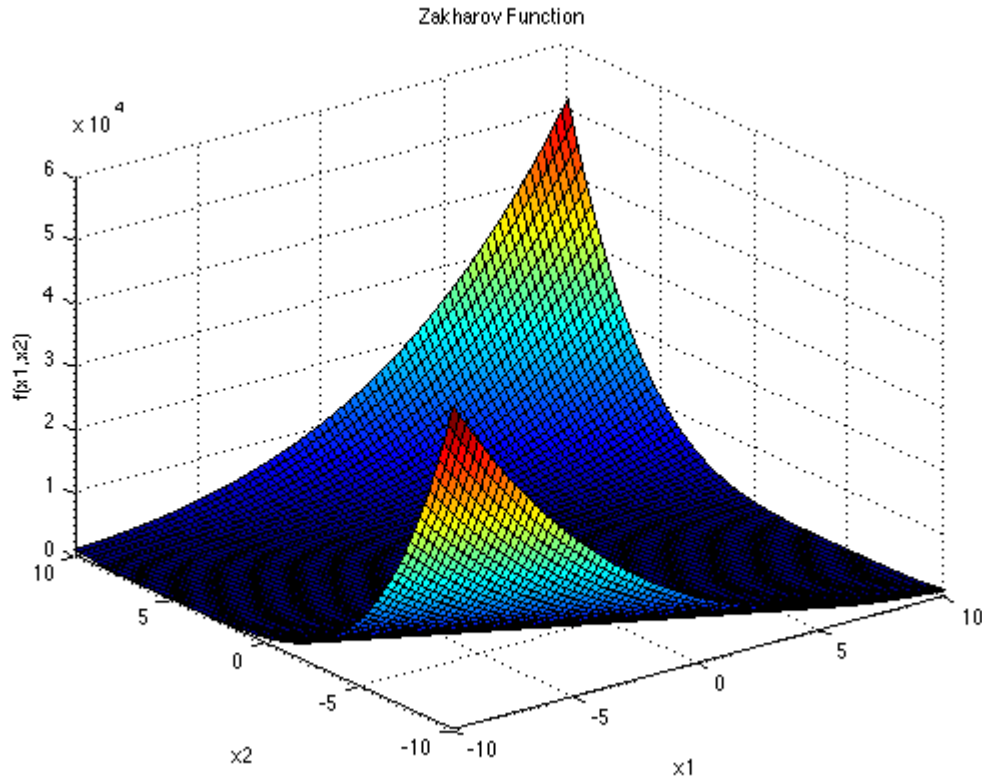
S.N o.	Initial Guess (x_0)	Exact Solution (x_c)	No of Iterations	Function Evaluations
1	[34.917,-12.531,-4.904,-12.184,-16.022,-14.848]	[6.00, 10.00, 12.00, 12.00, 10.00, 6.00]	94	2031
2	[4.773,34.708,19.506 ,8.94,20.979,8.546]	[6.00, 10.00, 12.00, 12.00, 10.00, 6.00]	91	1276
3	[-21.182,16.446,5.024,9.271,20.43,-4.697]	[6.00, 10.00, 12.00, 12.00, 10.00, 6.00]	80	1445
4	[33.828,-14.596,-3.667,-26.961,31.668,29.733]	[6.00, 10.00, 12.00, 12.00, 10.00, 6.00]	70	1661
5	[7.69,-16.669,27.04,15.6,8.481,28.27]	[6.00, 10.00, 12.00, 12.00, 10.00, 6.00]	80	1335
6	[-5.947,-35.486,19.852,-22.124,-26.043,15.649]	[6.00, 10.00, 12.00, 12.00, 10.00, 6.00]	84	1940
7	[-1.751,22.023,-25.612,-1.525,-18.749,-0.438]	[6.00, 10.00, 12.00, 12.00, 10.00, 6.00]	78	1796
8	[10.593,27.971,9.761,11.748,-15.148,35.443]	[6.00, 10.00, 12.00, 12.00, 10.00, 6.00]	74	1265
9	[20.063,-31.195,9.581,-3.332,14.272,-32.815]	[6.00, 10.00, 12.00, 12.00, 10.00, 6.00]	87	1857
10	[-35.364,26.751,-27.4,3.507,-33.661,2.699]	[6.00, 10.00, 12.00, 12.00, 10.00, 6.00]	95	2012

Problem 4: Trid Function (for 6 variables)



- The minima is found to match with analytical minima of $i(n + 1 - i)$
- The Objective function is $-\frac{n(n+4)(n-1)}{6}$ as expected
- Here $n = 6$, putting in expression, we get Optimal Objective Function Value of -50, which matches with numerical solution.

Problem 5: Zakharov Function (for 2 variables)



$$f(x) = \sum_{i=1}^2 x_i^2 + \left(\sum_{i=1}^n 0.5 \cdot i \cdot x_i \right)^2 + \left(\sum_{i=1}^2 0.5 \cdot i \cdot x_i \right)^4$$

Characteristics:

- Non-convex: Has a smooth surface with many different regions
- Multimodal: Large, flat regions around the global minimum.
- Separability: The function is not separable

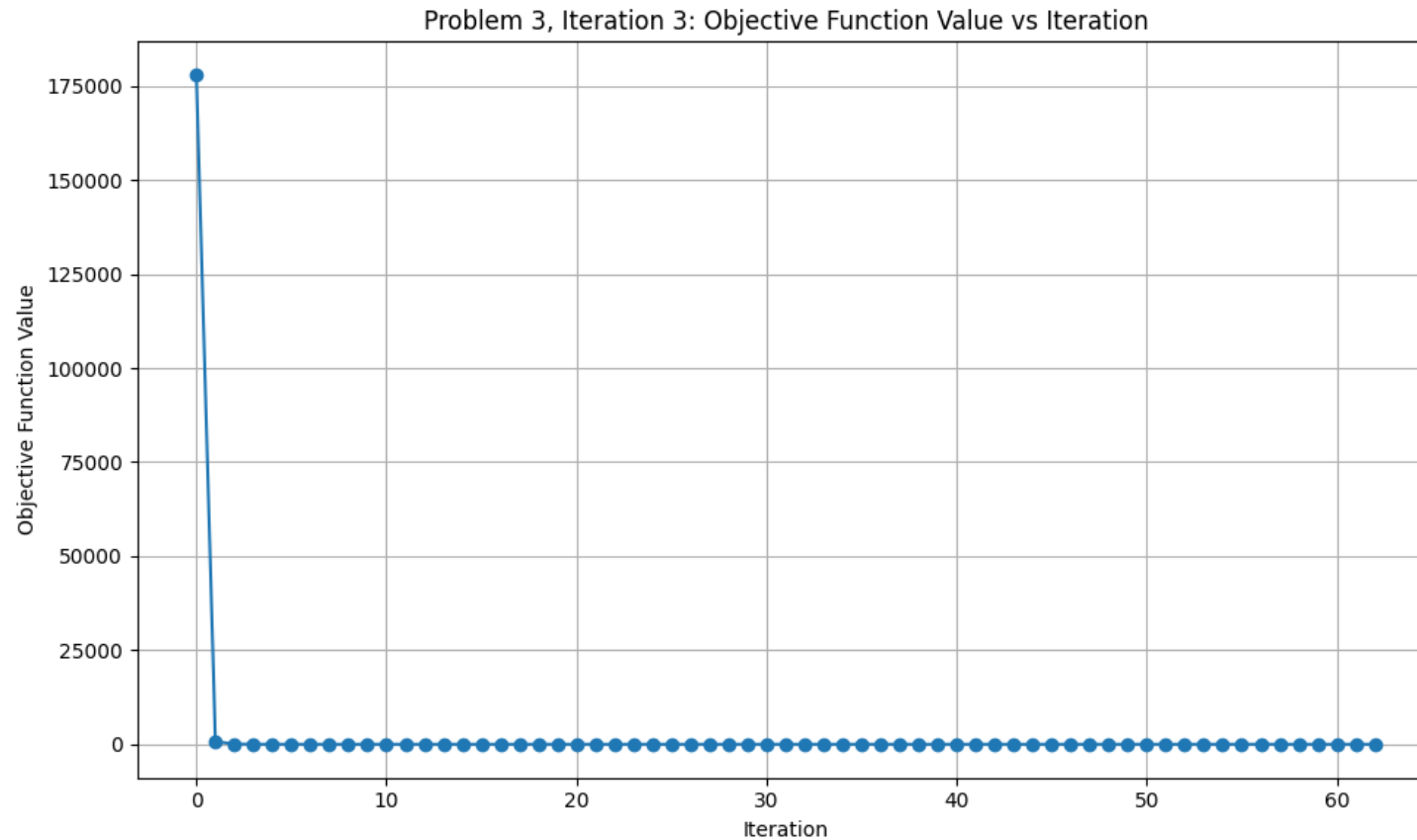
Global minimum:

- Occurs at $x = [0,0]$
- $f(x) = 0$

Problem 5: Zakharov Function (for 2 variables)

S.No.	Initial Guess (x_0)	Exact Solution (x_c)	No of Iterations	Function Evaluations
1	[5.622,8.127]	[0.00, 0.00]	4	88
2	[2.236,6.625]	[0.00, 0.00]	4	95
3	[5.695,-1.91]	[0.00, 0.00]	15	251
4	[2.461,-1.499]	[0.00, 0.00]	8	101
5	[-1.692,9.438]	[0.00, 0.00]	5	110
6	[6.66,-3.684]	[0.00, 0.00]	8	103
7	[-2.37,-4.09]	[0.00, 0.00]	3	81
8	[6.961,4.028]	[0.00, 0.00]	5	107
9	[5.051,5.971]	[0.00, 0.00]	5	100
10	[-0.327,6.671]	[0.00, 0.00]	5	102

Problem 5: Zakharov Function (for 2 variables)



- The minima is found to be when all variables are 0 as expected
- The Objective function is 0 as expected

Conclusion

- The conjugate gradient method is particularly effective for quadratic functions like Sum square and Trid, where it can converge faster in a finite number of steps. Also, good convergence has been observed in case of Zakharov function.
- For more complex function like Rosenbrock and Dixon-Price, the method is still effective but require more iteration and is highly sensitive to starting points.
- The issue with the Dixon Price function can because of many reasons:
 - Non Convex Function
 - Usage of bisection method (Any value based method like golden section search could have given better results)

Programming Project: Phase – 3

Constrained Optimisation using Bounding Phase, Bisection Method and Conjugate Gradient Method

ME – 609: Optimization Methods in Engineering

Group No: 18

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Agniva Ghosh - 210103009

Description

Bracket Operator Penalty Method

The Bracket Operator Penalty method is used for solving constrained optimization problems by transforming them into unconstrained problems. It does this by incorporating a penalty term that imposes a cost for violating constraints.

Method Steps:

1. Formulate the Penalty Function:

Given an optimization problem: $\min [f(x)]$ subject to $g(x) \leq 0$ the penalty function can be defined as: $P(x,r) = f(x) + r([\max(0, g(x))]^2$ where r is the penalty parameter that increases with iterations.

2. Parameter Selection

Penalty Parameter(r): Start with a small positive value and increase it iteratively to enforce constraints more strictly.

Initial Guess x_0 : An initial feasible or near-feasible solution.

Description

Conjugate Gradient Method

Method Steps:

1. Initialization:

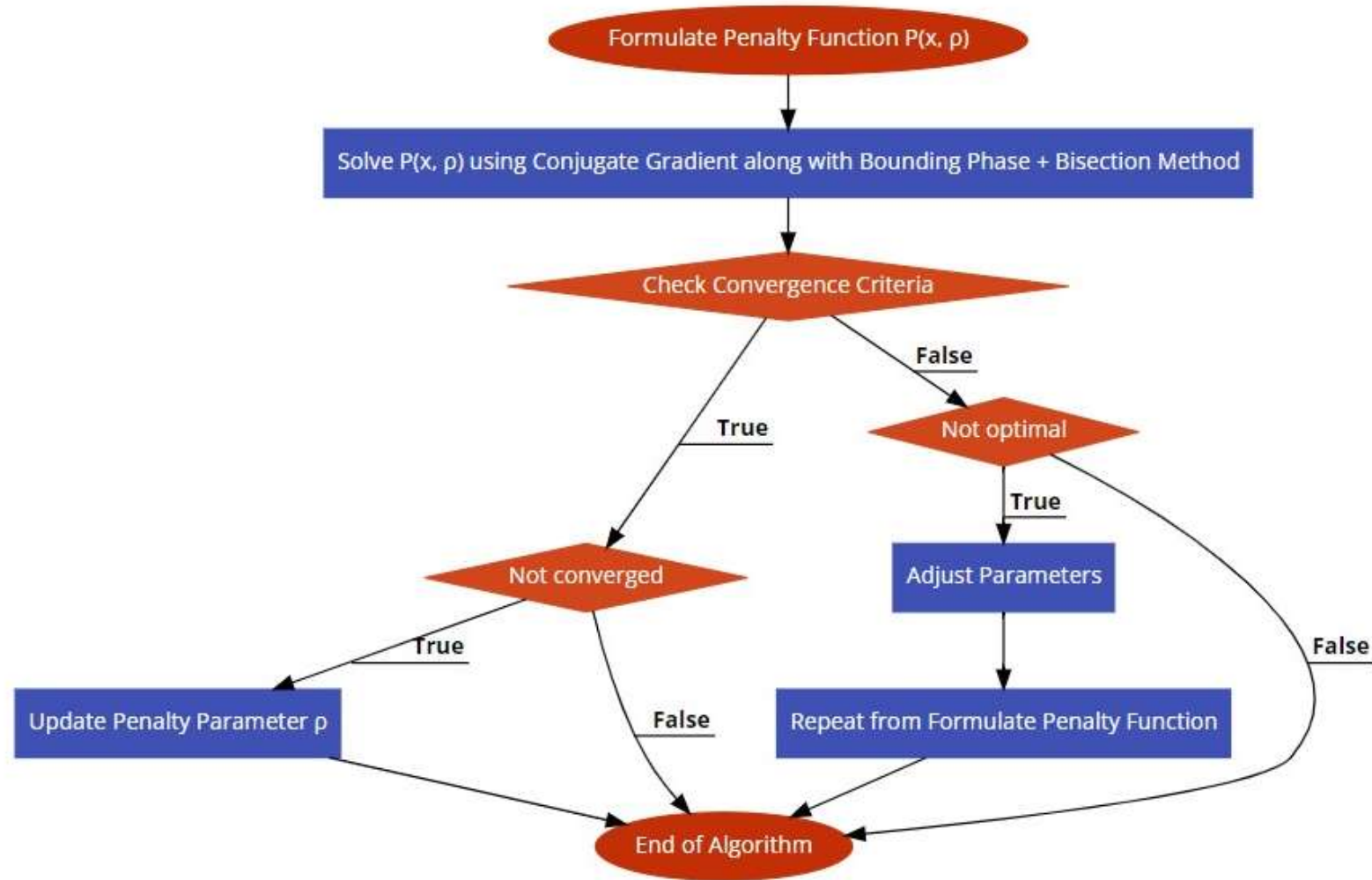
Start with an initial point x_0 and calculate the initial residual $d_0 = -\nabla P(x_0, r)$.

2. Iterate:

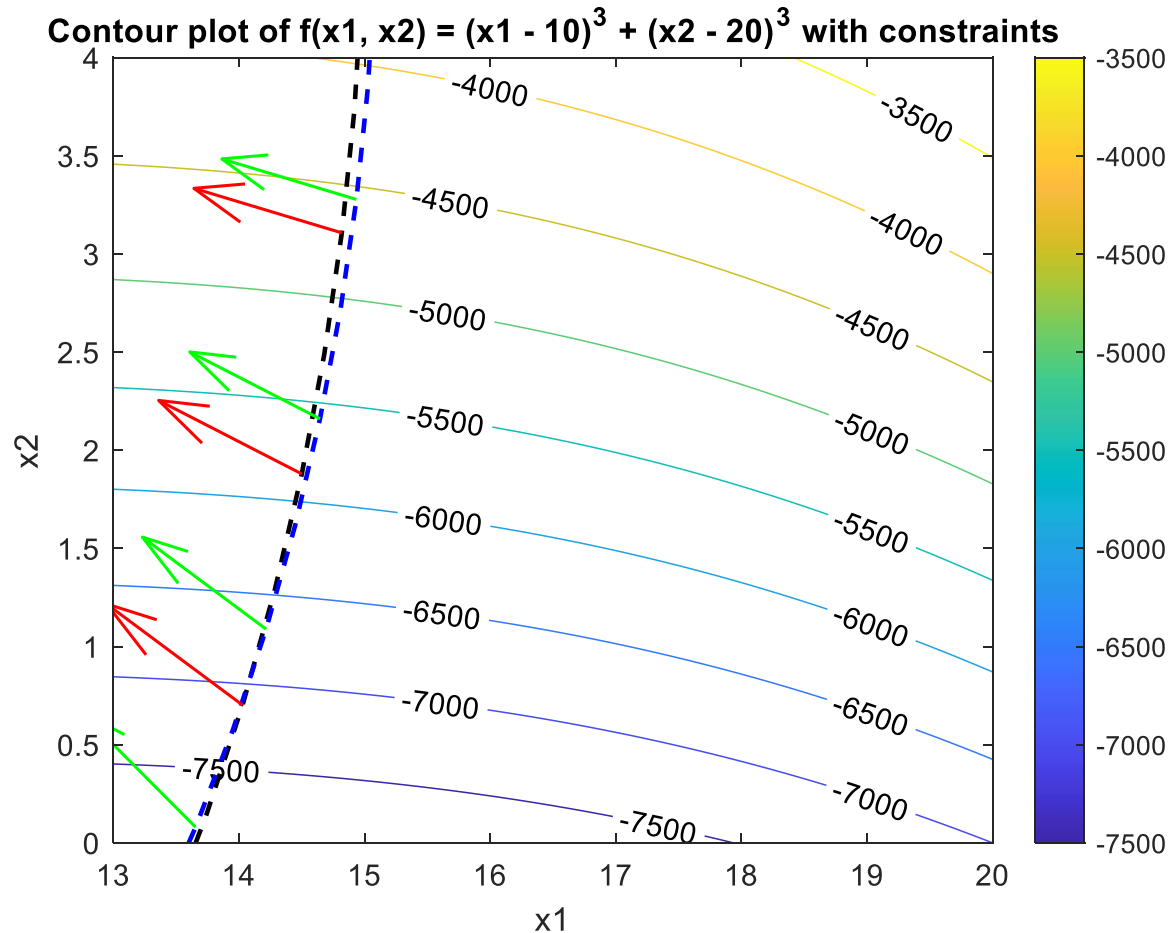
For each iteration

Compute the alpha size α_k using bounding phase + bisection method and update the solution $x_{k+1} = x_k + \alpha_k d_k$

Flowchart



Problem 1: $\min f(x) = (x_1 - 10)^3 + (x_2 - 20)^3$



$$\min f(x) = (x_1 - 10)^3 + (x_2 - 20)^3$$

Constraints for this problem:

$$g_1(x) = (x_1 - 5)^2 + (x_2 - 5)^2 - 100 \geq 0,$$

$$g_2(x) = (x_1 - 6)^2 + (x_2 - 5)^2 - 82.81 \leq 0,$$

$$13 \leq x_1 \leq 20, \quad 0 \leq x_2 \leq 4$$

Minima occurs at:

(14.095, 0.843)

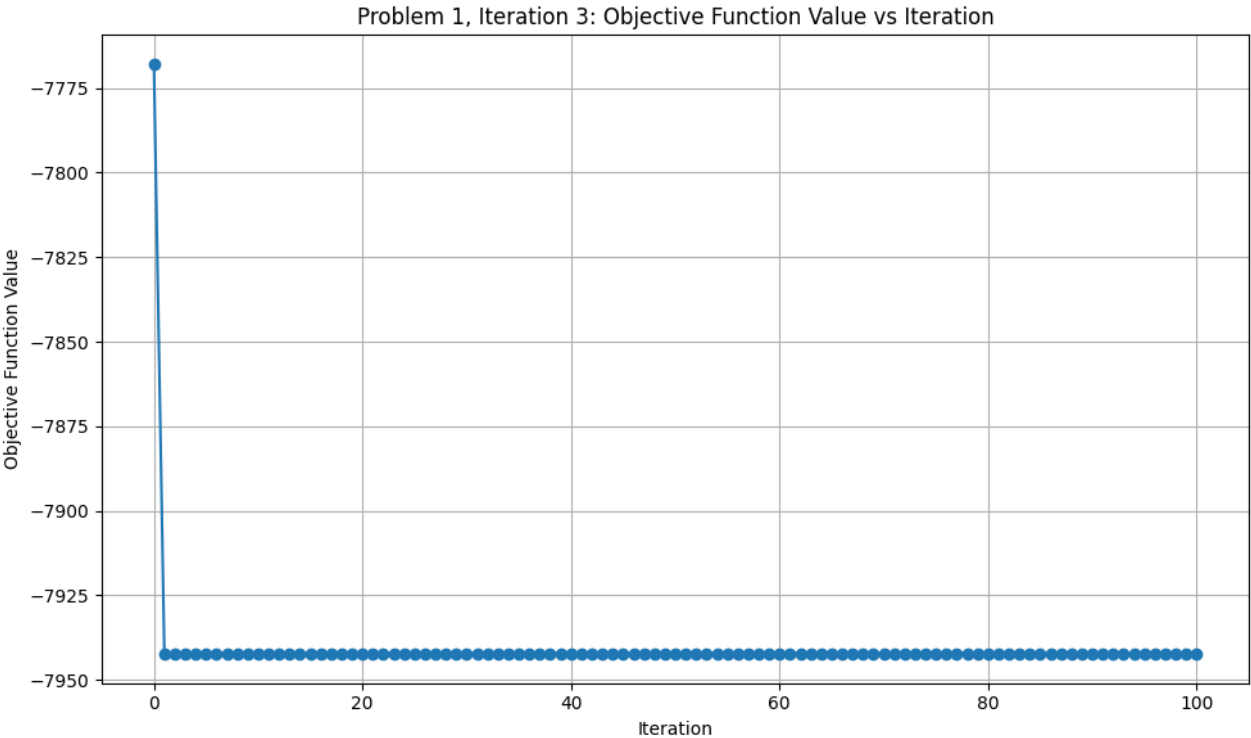
Optimum Value:

-6961.813

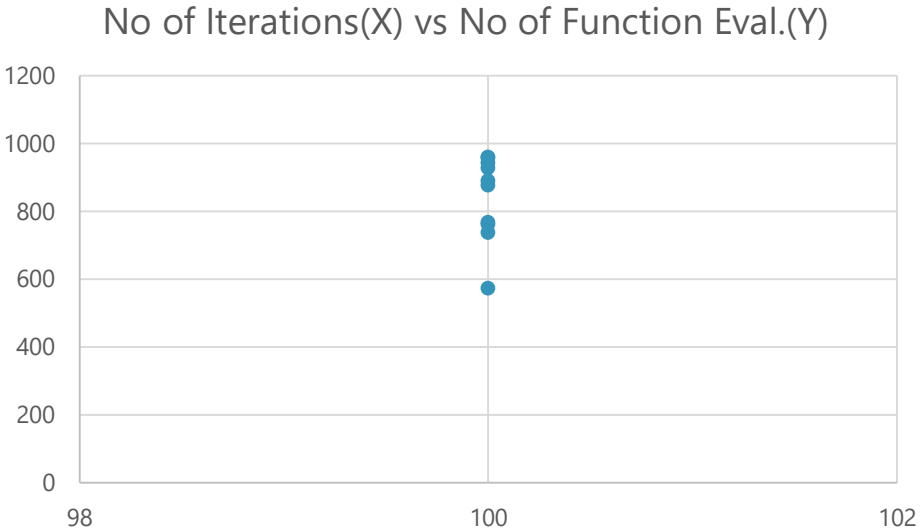
Problem 1: $\min f(x) = (x_1 - 10)^3 + (x_2 - 20)^3$

S.N o.	Initial Guess (x_0)	Obtained Solution (x_c)	Optimal Value	Number of Iterations	Function Evaluations
1	[14.561,0.551]	[14.321,0.001]	-7919.28	100	958
2	[14.183,0.494]	[14.091,0]	-7931.51	100	943
3	[13.394,0.133]	[13.411,0]	-7960.31	100	768
4	[13.733,0.236]	[13.736,0]	-7947.89	100	737
5	[13.062,0.903]	[13.425,0]	-7959.79	100	762
6	[13.728,0.939]	[13.895,0]	-7940.91	100	878
7	[13.036,0.488]	[13.194,0]	-7967.41	100	928
8	[14.298,0.838]	[14.164,0]	-7927.80	100	960
9	[13.315,0.971]	[13.647,0]	-7951.45	100	574
10	[13.881,0.668]	[13.852,0]	-7940.51	100	891

Problem 1: $\min f(x) = (x_1 - 10)^3 + (x_2 - 20)^3$



Parameters	Values
Best	-7967.41
Worst	-7919.29
Mean	-7944.69
Median	-7944.41
Std Dev	14.76557

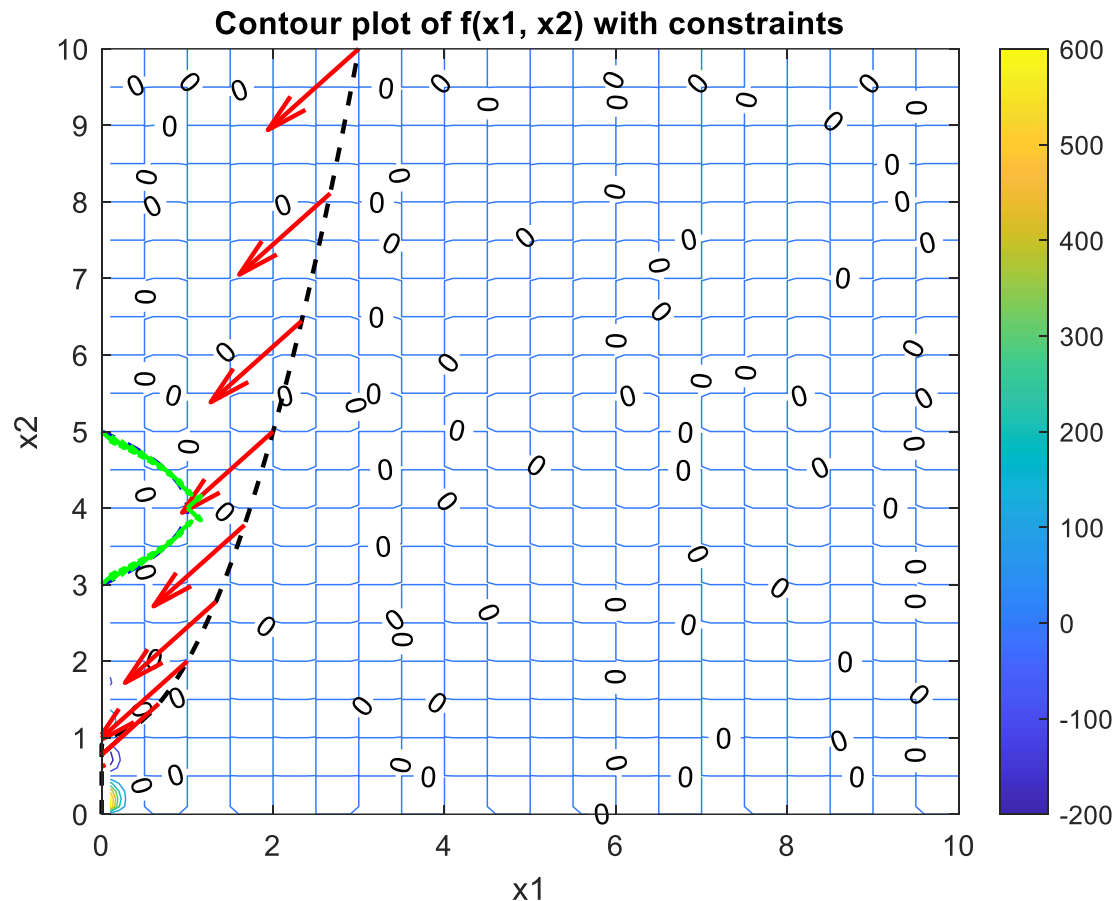


Problem 1: $\min f(x) = (x_1 - 10)^3 + (x_2 - 20)^3$

Observation:

- The numerically reported solution is quite far away from the expected solution
- We can see that while the x-coordinate results are pretty close, the y-coordinate results are not.
- Consequentially, as the objective function has cubic nature, small deviations from expected solution, will cause large changes in the solution.
- Moreover, the use of gradient based methods like Bisection method and Conjugate gradient method may not be suited to the objective function and the constraints given.
- A strange result is obtained wherein for all different initial values, the results converge after 100 iterations every time. This is not the case with problems 2 and 3.

Problem 2: $\max f(x) = \frac{\sin^3(2\pi x_1) \sin(2\pi x_2)}{x_1^3(x_1 + x_2)},$



$$\max f(x) = \frac{\sin^3(2\pi x_1) \sin(2\pi x_2)}{x_1^3(x_1 + x_2)},$$

Constraints for this problem:

$$g_1(x) = x_1^2 - x_2 + 1 \leq 0$$

$$g_2(x) = 1 - x_1 + (x_2 - 4)^2 \leq 0$$

$$0 \leq x_1 \leq 10, \quad 0 \leq x_2 \leq 10$$

Maxima occurs at:

(1.227, 4.245)

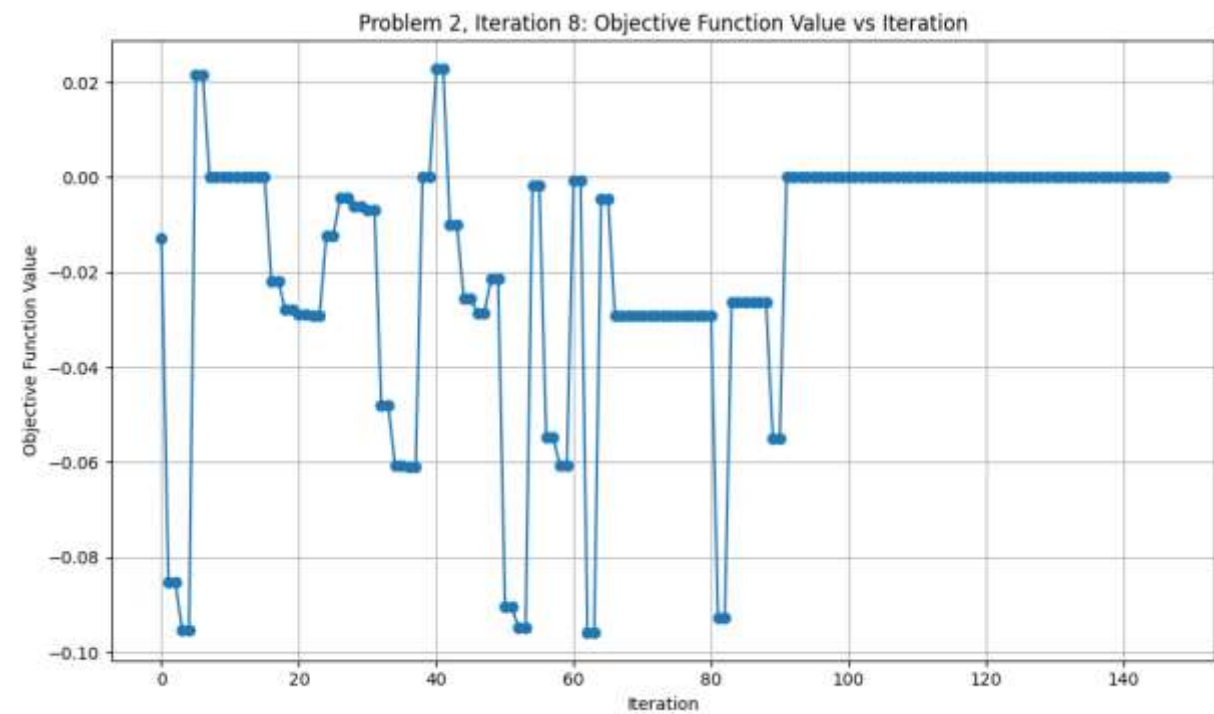
Optimum value:

0.985

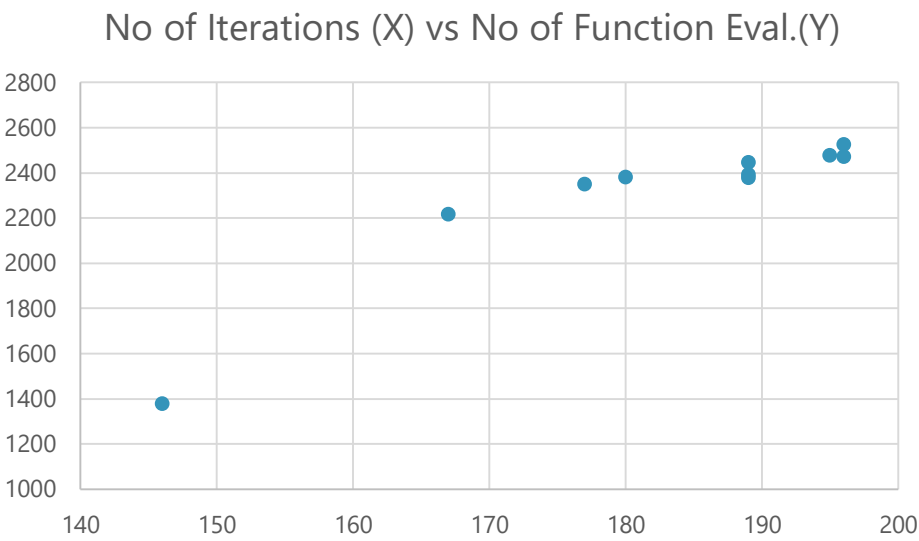
Problem 2: $\max f(x) = \frac{\sin^3(2\pi x_1) \sin(2\pi x_2)}{x_1^3(x_1 + x_2)},$

S.N o.	Initial Guess (x_0)	Obtained Solution (x_c)	Optimum Value	Number of Iterations	Function Evaluations
1	[1.899,1.924]	[1.227,4.245]	0.0958	177	2352
2	[1.878,3.289]	[1.227,4.245]	0.0958	196	2472
3	[1.165,4.351]	[1.228,4.245]	0.0958	189	2447
4	[1.495,4.992]	[1.227,4.242]	0.0958	195	2479
5	[1.013,4.245]	[1.227,4.245]	0.0958	189	2379
6	[1.446,2.501]	[1.227,4.234]	0.0953	180	2382
7	[1.192,0.261]	[1.227,4.245]	0.0958	189	2392
8	[1.390,4.135]	[1.227,4.245]	0.0958	146	1380
9	[1.965,3.417]	[1.227,4.245]	0.0958	196	2526
10	[1.731,3.026]	[1.227,4.245]	0.0958	167	2218

Problem 2: $\max f(x) = \frac{\sin^3(2\pi x_1) \sin(2\pi x_2)}{x_1^3(x_1 + x_2)},$



Parameters	Values
Best	0.09583
Worst	0.0954
Mean	0.09577
Median	0.09582
Std Dev	0.000127



Problem 2: $\max f(x) = \frac{\sin^3(2\pi x_1) \sin(2\pi x_2)}{x_1^3(x_1 + x_2)}$,

Observation:

- The numerically reported solution is very close to the expected solution
- We can see that the x-coordinate and the y-coordinate results are very accurate.
- Standard deviation value is very low implies that results are also very precise.
- We can see that although there is no direct link between no of function evaluations and no of iterations, it is evident that with increase in no of iterations, no of function evaluation increases. This is unlike problem 1.

Problem 3: $\min f(x) = x_1 + x_2 + x_3$

$$\min f(x) = x_1 + x_2 + x_3$$

Constraints for this problem:

$$g_1(x) = -1 + 0.0025(x_4 + x_6) \leq 0,$$

$$g_2(x) = -1 + 0.0025(-x_4 + x_5 + x_7) \leq 0,$$

$$g_3(x) = -1 + 0.01(-x_6 + x_8) \leq 0,$$

$$g_4(x) = 100x_1 - x_1x_6 + 833.33252x_4 - 83333.333 \leq 0,$$

$$g_5(x) = x_2x_4 - x_2x_7 - 1250x_4 + 1250x_5 \leq 0,$$

$$g_6(x) = x_3x_5 - x_3x_8 - 2500x_5 + 1250000 \leq 0,$$

$$100 \leq x_1 \leq 10000$$

$$1000 \leq x_i \leq 10000, i = 2, 3$$

$$10 \leq x_i \leq 1000, i = 4, 5, \dots, 8$$

Minima occurs at:

(579.317, 1359.943, 5110.071, 182.017,
295.598, 217.979, 286.416, 395.597)

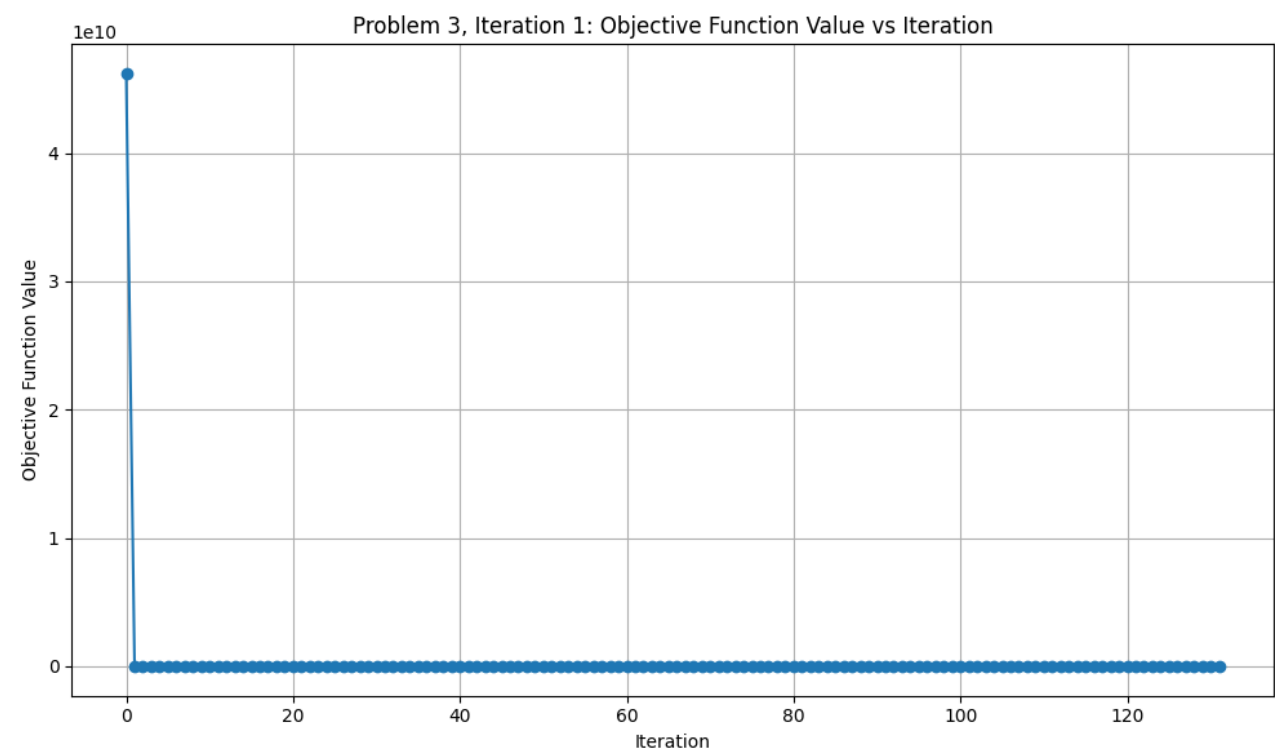
Optimum value:

7049.331

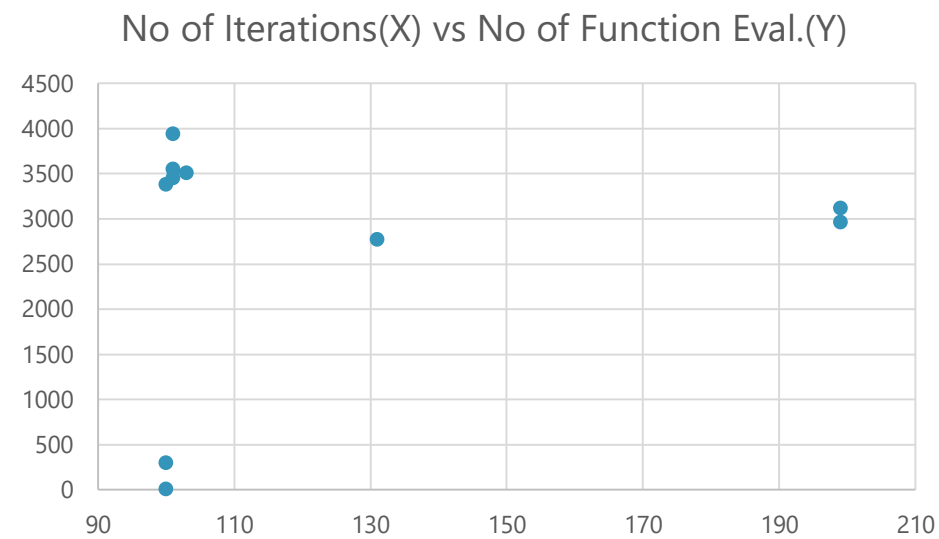
Problem 3: $\min f(x) = x_1 + x_2 + x_3$

S.No .	Initial Guess (x_0)	Obtained Solution (x_c)	Optimal Value	No of iterations	Function Evaluations
1	[691.18,1504.00,4114.42]	[662.87,1475.58,4088.05]	6226.515	131	2774
2	[477.22,1593.16,5519.73]	[477.22,1593.16,4088,05]	6158.44	100	9
3	[553.88,1480.12,5843.53]	[463.45,1389.69,5753.69]	7606.84	101	3942
4	[568.82,1576.23,4494.41]	[463.45,1470.86,4388.92]	6323.24	101	3450
5	[516.12,1580.12,4927.31]	[463.45,1573.92,4922.09]	6959.48	103	3512
6	[527.41,1626.62,4288.55]	[534.45,1626.62,4288.55]	6449.63	100	3386
7	[578.98,1243.72,4341.81]	[516.71,1188.63,4278.16]	5977.51	199	3118
8	[464.17,1563.93,5593.01]	[464.23,1563.93,5593.26]	7621.43	100	303
9	[644.84,1388.76,5202.29]	[596.89,1340.44,5154.72]	7092.06	199	2961
10	[659.40,1534.66,6047.51]	[463.45,1338.97,5852.08]	7654.49	101	3552

Problem 3: $\min f(x) = x_1 + x_2 + x_3$



Parameters	Values
Best	5977.511
Worst	7654.498
Mean	6806.968
Median	6704.559
Std Dev	627.1507



Problem 3: $\min f(x) = x_1 + x_2 + x_3$

Observation:

- The numerically reported solution is not close to the expected solution.
- We can see that both the x-coordinate and the y-coordinate results are not accurate.
- Standard deviation value is very high implies that results are also not precise.
- We can see that there is no direct link between no of function evaluations and no of iterations.
- Only for one iteration (Sl. No. 9) it is seen that the results are close to the expected solution.
- However, the mean and median results are not very far from the expected solution.
- Moreover, the use of gradient based methods like Bisection method and Conjugate gradient method may not be suited to the objective function and the constraints given.

Conclusion

- The use of Gradient based methods is suspected to be the root cause of inaccurate results. Any value based method (Golden-Section Search, Nelder-Mead Simplex Method) could have yielded better results than Bisection method and Conjugate Gradient method.
- No clear correlation between No. of Iterations and No. of Function Evaluations is observed, although in Problem 2, it is seen that increase in one value, tends to increase the other.
- The initial guess plays a big role in determining correct convergence. Many times, if the initial guess is far from the global minima, it is quite possible that the algorithm converges in the local minima, giving inaccurate results.
- Finally, if we can optimise the code, to take care of functions and constraints by using adaptive step sizes near locations of high function sensitivity/fluctuation, it can help in giving better results.